

Green transport fleet renewal using approximate dynamic programming: A case study in German heavy-duty road transportation

Jonas Winkelmann^{a,*}, Stefan Spinler^a, Thomas Neukirchen^b

^a Chair of Logistics Management, WHU – Otto Beisheim School of Management, Burgplatz 2, 56179 Vallendar, Germany

^b Duisburg Intermodal Terminal GmbH, Gaterweg 201, 47229 Duisburg, Germany

ARTICLE INFO

Keywords:

Transportation
Fleet renewal
Mobility transition
Decarbonization
Approximate dynamic programming
Simulation-based optimization

ABSTRACT

Governments and manufacturers are starting to enforce the European transport industry's transition to sustainable mobility. Meanwhile, transport companies have begun to set their own emissions goals. To achieve these sustainably, they must develop efficient policies to renew their fleets with alternative-fuel vehicles. However, since future trends in relevant parameters are highly uncertain, fleet managers struggle to make informed decisions. We formulate fleet renewal as a sequential optimization problem, considering multiple technologies and operational clusters. Vehicle purchase, sales, depreciation, fuel, carbon, and electric battery prices are modeled as stochastic variables. We propose approximate dynamic programming to calculate fleet renewal policies that achieve emissions goals while optimizing total costs of ownership. This approach is tested in a case study of a German logistics service provider. We investigate optimal timings of purchases and sales for a heavy-duty truck fleet, considering four drivetrain technologies. Our approach can guide decision making in various fleet renewal settings. By applying it to the case study, we derive important managerial implications. The mobility transition will significantly increase transport fleets' total cost of ownership. To minimize costs, companies should not move prematurely to low-emissions technologies, but hold vehicles for as long as possible to benefit from fewer purchases and sinking prices. The optimal policy depends on the distance driven. For short-distance operations, diesel trucks will remain the dominant technology in the next years, but will be replaced by battery electric trucks in the medium term. In the far future, trucks powered by electricity and hydrogen will be equally important.

1. Introduction

As part of the Paris Agreement and the European Green Deal, Germany has committed to introducing national measures to counter climate change and limit the global warming to 1.5 degrees Celsius relative to pre-industrial levels (United Nations, 2015; European Commission, 2019). In 2021, the German government passed a national climate protection law, which strengthened its decarbonization ambitions. Accordingly, the target is to reduce cross-sectoral emissions to 35% of 1990 emissions levels by 2030, 12% by 2040, and net zero by 2045 (Bundesministerium der Justiz, 2021). A major challenge is to reduce carbon emissions in the transport sector. In 2020, the transport sector is said to have accounted for 22% of total emissions from EU countries, equivalent to

* Corresponding author.

E-mail addresses: jonas.winkelmann@whu.edu (J. Winkelmann), stefan.spinler@whu.edu (S. Spinler), thomas.neukirchen@dit-duisburg.de (T. Neukirchen).

<https://doi.org/10.1016/j.tre.2024.103547>

Received 4 October 2023; Received in revised form 17 April 2024; Accepted 19 April 2024

Available online 29 April 2024

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roughly 729 megatons CO₂e. Sadly, this number is overly positive, as the COVID-19 pandemic temporarily reduced transportation compared to previous years (e.g., 835 megatons CO₂e in 2019) (European Environment Agency, 2021).

From a company perspective, measures to reduce transport emissions may be more beneficial than measures in other areas. Such measures are available at comparatively low cost and promise relatively high savings and short payback periods (Blanco, 2022). However, companies are still at an early stage of moving toward sustainability, and often lack the capabilities to design and run sustainable transport operations (McKinnon and Petersen, 2021). Operations management research needs to provide guidance on designing efficient strategies to lower companies' environmental impact (Drake and Spinler, 2013). Shifting to cleaner transport modes and alternative energy forms are among the most cost-effective measures to achieve this (McKinnon and Petersen, 2021). Consequently, efficient policies must be developed to help transport companies successively renew their fleets with alternative-fuel vehicles (AFV). For example, internal combustion engine vehicles powered by diesel (ICEV) should be replaced with liquified natural gas vehicles (LNGV), battery electric vehicles (BEV), or fuel cell electric vehicles (FCEV).

The problem of replacing fleet vehicles has received extensive academic attention, particularly in mathematical optimization. In most of such research, exact optimization algorithms are applied to derive how and when a particular action (e.g., buy, sell, lease, retrofit) should be carried out (Figliozzi et al., 2011; Kleindorfer et al., 2012; Pantuso et al., 2016; Pelletier et al., 2019). However, these traditional approaches are reaching the limit of their ability under the circumstances of the forthcoming mobility transition, for three main reasons known as the “curses of dimensionality” (Powell, 2011). These are the high degree of uncertainty (e.g., in vehicle prices), long time horizons requiring sequential decisions (e.g., until 2050), and choices between multiple technologies, affecting the dimensions of the decision variables and the number of times a solution must be calculated. The underlying optimization problem grows significantly in complexity and size, eventually becoming computationally intractable for exact approaches. In contrast, approximate approaches relying on reinforcement learning techniques are gaining popularity to effectively overcome the “curses of dimensionality” in logistics and supply chain management applications (Yan et al., 2022). Recent studies prove they offer a powerful toolset capable of solving these large and complex problems (see Ding et al. (2023), Chow et al. (2024), Stranieri et al. (2024)). Despite that, they are yet rarely applied to derive decision policies considering environmental or sustainability factors (Yan et al., 2022).

This paper presents a methodology to find valuable approximate solutions to the problem of renewing transport fleets with AFVs. We formulate fleet renewal as a sequential stochastic decision problem considering multiple technologies and vehicle clusters based on operational characteristics. The target function minimizes the total cost of ownership (TCO) across all periods. The TCO includes all costs (and revenues) incurred through purchasing, operating, and selling the vehicles over their entire lifetime (see Ellram and Siferd (1993)). In the meantime, emissions goals must be achieved. An approximate dynamic programming (ADP) approach solves the problem by combining selected methods from simulation, optimization, and reinforcement learning. The program moves forward through a specified timeframe, calculating the state of the transport fleet and the development of relevant parameters. Selected cost variables are modeled stochastically due to their high degree of uncertainty. This concerns vehicle purchase prices, sales prices, depreciation, fuel prices, carbon prices, and electric battery prices. In each period, the program optimizes decisions to buy and sell transport vehicles across multiple technologies using a metaheuristic (dual annealing). A value function approximation (VFA) of future costs is implemented in the form of a linear parametric model that continuously learns the value of being in a particular state, taking account of the stochastic nature of the parameters. Consequently, with a growing number of runs, the program can learn (near-)optimal solutions to the large-scale optimization problem.

The ADP results can be used to derive meaningful policies and managerial insights for companies aiming to renew their transport fleets with AFVs. Ideally, the model would be used iteratively (e.g., every year, every second year, etc.) while continuously updating forecasts and assumptions to leverage the latest and most reliable information. As a result of the stochastic modeling and the applied methodology, the calculated policies constitute decision rules considering certain problem states and parameter developments. Thus, companies can understand their uncertain circumstances and react to them, giving them confidence to meet their emission goals cost-efficiently. The methodology can be applied to various circumstances in road transportation, including different transport fleet types (e.g., trucks, vans), different operational characteristics (e.g., tour lengths, shift models), different competing technologies (e.g., ICEV, BEV, FCEV), and different emission reduction targets. As part of this research, we apply our approach to a real-world case study of an anonymous German logistics service provider (LSP). The LSP operates heavy-duty trucks on tours around regional hubs, covering short to medium annual distances (i.e., 20,000 to 170,000 km). We evaluate the adoption of four competing technologies: ICEV, LNGV, BEV, and FCEV trucks. The circumstances of this fleet renewal problem are very common for LSPs in the trucking sector. For instance, major companies like DB Schenker and the Deutsche Post DHL Group announced comparable emissions targets and develop strategies to continuously sell trucks with conventional fuels and purchase alternative fuel trucks instead (The Climate Choice UG, 2022; Deutsche Post DHL Group, 2022). In this context, our results are leveraged to answer two key questions: which alternative technologies should transport companies invest in during the mobility transition, and when is the right time to buy and sell these technologies?

This paper is structured as follows. Section 2 provides an overview of the literature. In Section 3, we explain our methodology. The problem is formulated as a sequential stochastic problem, and the components of the ADP approach are explained. The specifics of the LSP case study are described in Section 4. Section 5 presents our computational experiments to validate the approach and produce numerical results. In Section 6, we discuss the managerial implications of our findings and their relevance to the research community. Section 7 summarizes our conclusions.

2. Literature review

The inherent optimization problem for companies seeking to renew or replace their transport fleets has been studied from various perspectives using different terminology, such as “fleet replacement problem,” “equipment replacement problem,” and “asset acquisition problem.” We understand fleet renewal to mean the continuous replacement of transport vehicles over a specific time horizon. The optimized decisions can be a company’s vehicle purchases and sales or similar actions like leasing or retrofitting. Naturally, these operators may be subject to specific conditions. Our study also accounts for multiple technologies, vehicle clusters, and random parameters. Together with the number of time steps, these problem characteristics drive the fleet renewal problem’s scale and complexity. A vast spectrum of methodologies has been applied to derive meaningful decision support (Wang et al., 2013; Lim et al., 2015; Pantuso et al., 2016; Kuppusamy et al., 2017). This paper focuses specifically on the application of ADP, which is a simulation-based optimization approach leveraging elements from mathematical programming and classical simulation (Powell, 2011). Therefore, this literature review focuses on two relevant areas: research applying a related methodology (e.g., mathematical programming, stochastic dynamic programming) to the fleet replacement problem, and research that explicitly proposes ADP in the broader area of fleet replacement, equipment replacement, or asset acquisition problems. We review the selected papers regarding their problem setup dimensions (i.e., deterministic/stochastic, number of considered random factors, time periods, technologies, clusters) and their algorithm design (i.e., methodology, exact/approximate). Table 1 summarizes the results.

Classical mathematical programming calculates exact solutions and has predominantly been applied to deterministic problems (i.e., no random factors are modeled). Such publications approach uncertainty only by evaluating alternative scenarios or calculating descriptive risk metrics. In this group, various authors formulate the problem as an integer (linear) program and solve it using a range of commercial solvers. The studied problems involve 15 to 31 time periods and two to four technologies but no clusters (Alp et al., 2022; Figliozzi, 2013; Pelletier et al., 2019). Similarly, formulations of the deterministic problem as mixed-integer linear programs are widespread. Again, applying commercial solvers is the most common solution approach. Similar problem dimensions can be observed. They feature ten to 30 periods, one to three technologies, and no clusters (Abdi and Taghipour, 2018; Ahani et al., 2016; Figliozzi et al., 2011).

Other contributions, which still apply exact optimization algorithms, deliberately model issues stochastically, referring to these as sequential stochastic problems, multi-stage stochastic problems, or Markov decision processes. Some researchers also apply commercial solvers to these considerably larger problems. This approach seems to be popular in naval shipping, as several authors leverage it to renew fossil-fueled (i.e., one technology only) vessel fleets with uncertain demand, freight rates, charter rates, operations costs, capacities, or vessel purchase and sales prices. None of the reviewed papers regard fleet clusters (Arslan and Papageorgiou, 2017; Bakkehaug et al., 2014; Pantuso et al., 2014, 2016). Regarding the other problem dimensions, Arslan and Papageorgiou (2017) study two of the specified random factors and 12 time periods. Bakkehaug et al. (2014) consider 11 random factors and up to eight periods. Pantuso et al. (2016) model nine stochastic parameters and six periods. Although the limitation to a single energy source reduces the problem size, they highlight drastically increasing computation times when the number of variables is raised. Ansariipoor et al. (2014, 2016) also solve their sequential stochastic problem using a commercial solver. They optimize the fleet of a UK-based company by incorporating three technologies (fossil fuel, hybrid, electric) and six stochastic parameters (CO_2 price, fuel prices, mileage, fuel consumption) over four time steps. Additionally, they propose a risk factor (Recursive Expected Conditional Value at Risk) that depends on the degree of uncertainty and is integrated into the target function. The more recent paper (Ansariipoor et al. (2016)) also considers dividing the fleet into three to five clusters to improve the optimization outcome. Extending their earlier studies, Ansariipoor and Oliveira (2018) apply the real options methodology to the decision between diesel, petrol, and electric vehicles (i.e., three technologies). They consider various leasing contract options, the aforementioned risk measure, no clusters, and one additional random parameter (BEV lease costs). The new parameter is modeled with an Ornstein–Uhlenbeck process, and different technological development speeds are tested as part of a sensitivity analysis. Except for that, the problem dimensions remain unchanged. Stålhane et al. (2019) develop a two-stage stochastic programming approach to plan single-technology maintenance vessel fleets used at offshore wind parks. The study does not indicate the number of modeled periods, and fleet clusters are not mentioned. In the first stage, the model decides what vessels to charter and which hubs to use. In the second stage, the vessels are deployed to support the maintenance tasks. While two uncertain parameters are considered (weather conditions and failure occurrence), they are only relevant for the second stage and not for the first stage, where the fleet is composed. The problem is decomposed and solved with a matheuristic approach. Stochastic dynamic programming is another very popular class of exact methods used for stochastic fleet renewal problems. Fan et al. (2014) apply the technique to optimize the vehicle replacement in the fleet of the Texas Department of Transportation. They consider one stochastic factor (vehicle utilization), only one technology, and no clusters across 20 periods. Hsu et al. (2011) propose a similar approach to support airlines in their decisions to purchase, lease, or discard planes in a timeframe of eight periods. While no clusters and only one technology is regarded, the study accounts for one random parameter. The uncertain demand is modeled by combining the Grey topological forecast method and Markov chains. In contrast to these papers, Kleindorfer et al. (2012) develop a decision model for the French postal service to support decisions on adopting electric over conventional vehicles (i.e., two technologies). Sixty time periods are modeled in total. They take account of uncertain fuel and battery prices and model the two random parameters as a Brownian motion with drift and an Ornstein–Uhlenbeck process. Neboian and Spinler (2015) expand on this paper by giving the additional option to breach leasing contracts. Regarding the problem dimensions, only the number of considered time steps changes (20 periods).

Fleet replacement problems may quickly become computationally intractable as the number of decision variables or the amount of required and often stochastic information increases (Powell, 2011). Approaches that calculate exact solutions are therefore limited to relatively small-sized problems. In other words, the approaches discussed above cannot be used to provide solutions to large instances

Table 1
Overview of reviewed literature with regards to problem setup and algorithm choice.

Reference	Problem setup					Algorithm design	
	Uncer-tainty	Random factors	Time periods	Techno-logies	Clusters	Approach	Accuracy
Alp et al. (2022)	determ.	–	15	2	1	CS	exact
Figliozzi (2013)	determ.	–	30	2	1	CS	exact
Pelletier et al. (2019)	determ.	–	31	4	1	CS	exact
Abdi and Taghipour (2018)	determ.	–	10	1	1	CS	exact
Ahani et al. (2016)	determ.	–	30	2	1	CS	exact
Figliozzi et al. (2011)	determ.	–	14	3	1	CS	exact
Arslan and Papageorgiou (2017)	stoch.	2	12	1	1	CS	exact
Bakkehaug et al. (2014)	stoch.	11	1–8	1	1	CS	exact
Pantuso et al. (2016)	stoch.	9	6	1	1	CS	exact
Ansariipoor et al. (2014)	stoch.	6	4	3	1	CS	exact
Ansariipoor et al. (2016)	stoch.	6	4	3	3–5	CS	exact
Ansariipoor and Oliveira (2018)	stoch.	7	4	3	1	CS	exact
Stålhane et al. (2019)	stoch.	2	–	1	1	SP, H	exact
Fan et al. (2014)	stoch.	1	20	1	1	SDP	exact
Hsu et al. (2011)	stoch.	1	8	1	1	SDP	exact
Kleindorfer et al. (2012)	stoch.	2	60	2	1	SDP	exact
Neboian and Spinler (2015)	stoch.	2	20	2	1	SDP	exact
van Duin et al. (2013)	determ.	–	1	2	1	H	approx.
Zhang et al. (2021)	determ.	–	20	1	1	H, MH	approx.
Turan et al. (2020)	stoch.	10	10	1	1	MH, S	approx.
Turan et al. (2022)	stoch.	10	80	1	1	MH, S	approx.
Ali et al. (2023)	stoch.	6	10	1	1	MH, S	approx.
Sadeghpour et al. (2019)	stoch.	3	5	–	1	ADP	approx.
Stasko and Oliver Gao (2012)	stoch.	1	25	1	1	ADP	approx.
This paper	stoch.	13	29	4	4	ADP	approx.

CS - Commercial solver, SP - Stochastic programming, SDP - Stochastic dynamic programming,
H - Heuristic, MH - Metaheuristic, S - Simulation, ADP - Approximate dynamic programming

of the fleet renewal problem with AFVs. To illustrate this issue, we provide example calculations of the required computational time to solve such a problem in [Appendix A](#). In contrast, approximate methods are not typically applied to find the best solution to an optimization problem. Rather, they are used to find sufficiently good solutions much more quickly, which makes them valuable for large-scale problems ([Powell, 2022](#)). Only a few authors apply approximate algorithms to a deterministic problem setting (i.e., no random factors), because such problems can usually be solved with exact methods. For example, [van Duin et al. \(2013\)](#) study the integration of electric vehicles into an urban transportation fleet (i.e., two technologies) in Amsterdam and evaluate operational uncertainties as part of various scenarios. They use a sequential insertion heuristic to solve the problem and only model a single time step without considering multiple clusters. Moreover, [Zhang et al. \(2021\)](#) use a two-stage heuristic approach based on continuous-time approximation and tabu search to solve the fleet replacement problem. The first stage generates constraints that reduce the solution space and guarantee a solution close to optimality. On this basis, the second stage decides when to purchase and retire vehicles. The authors do not consider AFVs (i.e., one technology) or clusters but model a medium number of time periods (20).

Our core theme is using approximate optimization algorithms for stochastic problems generated through simulation. [Turan et al. \(2020\)](#) use the genetic algorithm metaheuristic to find promising sets of renewal strategies for a naval shipping fleet. They then evaluate these strategies by leveraging system dynamics and discrete event simulation. Their research considers ten different resource and operational uncertainties, but only one technology is evaluated. The study covers ten decision periods and does not account for clusters. In a second study, [Turan et al. \(2022\)](#) formulate a problem combining workforce planning and fleet renewal issues in a military setting. They leverage a genetic algorithm to create solutions and assess them with a system dynamics model. The ten considered uncertainties are exclusively related to the specific context, and only a single drivetrain technology and no clusters are evaluated. However, they consider the largest number of time steps (80). Extending these previous publications, ([Ali et al., 2023](#)) apply a similar approach to provide policies for fleet modernization with the goal of maximizing the deployment of military assets while minimizing costs. Again, the genetic algorithm is used in combination with a (capability) simulation model to generate and improve initial solutions and evaluate their quality. A selection of six stochastic parameters is considered that predominantly describe asset availability and status. Apart from that, ten periods, a single technology, and no clusters are modeled. [Sadeghpour et al. \(2019\)](#) apply an approach similar to ours. They use ADP solved by a genetic algorithm metaheuristic to tackle a large-scale equipment replacement problem for transformers in an energy supply company. The authors consider three uncertainties related to errors caused by human, instrumental, and environmental factors. The study covers five periods and no clusters. However, to date, applications of ADP to the fleet renewal problem as in our study are scarce. Only [Stasko and Oliver Gao \(2012\)](#) present both a comparable problem setting and a similar methodology. They apply ADP to support purchase, resale, and retrofit decisions in a truck fleet over 25 time steps. They consider stochastically appearing vehicle breakdowns (i.e., one random parameter) but only examine diesel vehicles (i.e., one technology) without considering clusters.

[Table 1](#) summarizes the reviewed literature with regard to the studied problem setup and the applied algorithms. Concerning the problem setup, all papers deal with limited problem sizes. They primarily focus only on two out of the four factors driving problem

complexity (i.e., the number of random factors, time periods, technologies, and clusters). Instead, during this paper's case study, we deal with high numbers on all four dimensions. In comparison, we study the most random factors (13), the most technologies (four), and a very high number of time periods (29) and clusters (four). Regarding the choice of algorithms, the application of approximate algorithms is still scarce (seven out of 24). This is controversial, as they are essential to solve large instances of the fleet renewal problem with AFVs. At the same time, the approximate algorithms that have been applied to stochastic problems did not focus on the examination of multiple technologies and did not account for clusters of vehicles.

In summary, this paper addresses a clear research gap. First, we propose a generic formulation of transport fleet renewal as a sequential stochastic problem. This features multiple technologies, operational clusters, and specific stochastic parameters. The problem is also constrained by individual emissions targets, so its solutions represent policies to accomplish the mobility transition cost-efficiently. Second, we develop an ADP approach to solve this large-scale problem approximately. Third, we apply the approach to a case study of a leading German LSP. No previous study has evaluated four competing technologies in a setting with stochastic purchase, sales, depreciation, fuel, carbon, and battery prices across a timeframe of comparable size for a real case. As a result, we derive unique managerial insights for transport companies seeking to renew their transport fleets with AFVs to meet their emissions targets.

3. Methodology

3.1. Generic problem description

We understand fleet renewal to mean the continuous replacement of transport vehicles over a specific time horizon, expressed through a company's vehicle purchases and sales under specific conditions. Transport companies strive to minimize their fleet's TCO over a certain time horizon. The TCO comprises all cost types ultimately resulting from a purchase decision. It helps companies look beyond the purchase price and consider the true cost of a good (or a transport service) throughout its entire lifecycle. For example, costs relating to operating or salvaging an asset are also regarded. Ultimately, considering the TCO leads to a better understanding of cost structures and beneficial purchase decisions (Ellram and Siferd, 1993). Fleet renewal is naturally regarded as sequential because decisions about purchases and sales are taken in sequenced time steps or periods $t \in \mathcal{T} = \{1, 2, \dots, T\}$. Due to the randomness of certain cost parameters, the problem is further defined stochastically. Specifically, vehicle purchase prices, sales prices, depreciation, energy prices, carbon prices, and battery prices are modeled as stochastic processes. This choice of random parameters is based on the current academic literature and discussions with practitioners ranging from logistics service providers to vehicle manufacturers. They are also the ones that matter for the management of the fleet operator in the case study. Note that we deliberately do not model any additional political instruments (e.g., ICEV tolls, street bans) and their uncertainty. We assume the political environment to be deterministic and stable, reflecting its status quo. Through modeling the mentioned stochastic parameters, we already account for some of these political effects (e.g., different carbon price levels), and adding additional instruments could lead us to double-count and oversize them. Instead, this study focuses on understanding how a typical transport company would react to certain cost developments. Considering a policymaker's perspective, applying different political instruments to influence companies' fleet renewal with AFVs shall be subject to future studies.

In this study, the fleet renewal problem also involves alternative technologies $e \in \mathcal{E} = \{1, 2, \dots, E\}$ and different vehicle clusters $v \in \mathcal{V} = \{1, 2, \dots, V\}$. Accounting for multiple drivetrain technologies enables their comparison and evaluation while solving the fleet renewal problem. The results can be used to inform companies on cost-efficient ways to undertake their mobility transition. Moreover, it may make sense to group several fleet vehicles based on their operational characteristics and carry out separate optimizations for each. We call these groups vehicle clusters. Relevant operational metrics could be, for example, the average distance driven or the average load. Vehicles might share similar metrics values within the groups, but the values might significantly differ between the clusters. In this case, divergent decarbonization policies could be preferable for each group. For example, long-distance and heavy-load clusters might have issues decarbonizing, as certain AFVs' mileage or weight constraints are violated. Furthermore, long-distance and short-distance clusters might have different beneficial technologies. We illustrate this in Fig. 1. From a company perspective, considering vehicle clusters might lead to valuable prioritization strategies. Selected clusters may decarbonize faster than others, driving the achievement of aggregate company constraints such as CO₂e targets. Additionally, the vehicles' age $y \in \mathcal{Y} = \{0, 1, \dots, Y\}$ and mileage $m \in \mathcal{M} = \{0, 1, \dots, M\}$ need to be accounted for. These parameters determine the vehicles' retirement from the fleet based on their maximum values Y and M . They also influence the calculation of certain cost types (e.g., depreciation, sales price).

3.2. Definition of the sequential stochastic problem dimensions

In this section, we present our formulation of fleet renewal as a sequential stochastic problem (or Markov decision process). In a Markov decision process, the information describing the state of the fleet in period t is captured in the state variable S_t . The decisions regarding purchases and sales are contained in the action vector a_t from the total action space A_t , which depends on the state S_t . Following the generic problem formulation, a transport company aims to pursue actions a_t that minimize a TCO function $C_t(S_t, a_t)$ over time. Due to the involved stochasticity, we are rather interested in finding a decision rule or policy function π from the set of all potential policies Π that returns a beneficial action a_t given a particular state S_t . The optimization problem can be written as:

$$\min_{\pi \in \Pi} \mathbb{E}^{\pi} \left\{ \sum_{t=0}^T \gamma^t C_t^{\pi}(S_t, A_t^{\pi}(S_t)) \right\}, \quad (1)$$

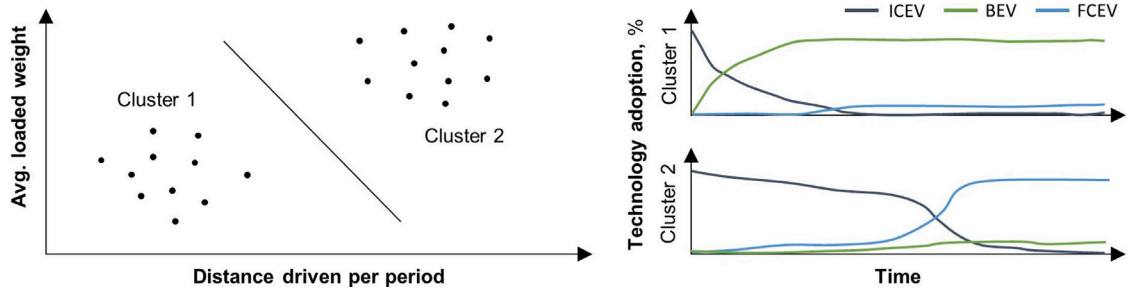


Fig. 1. Illustration of potential vehicle clusters (left) and their hypothetical decarbonization policies (right).

where γ is a discount factor (see Powell (2011)). Note that yielding such policies π is one of the key strengths of the approach compared to solving a deterministic problem version. Transport companies face severe uncertainty due to the forthcoming mobility transition. Solving a deterministic version proposes a single solution for an assumed future, which will likely turn out differently. In contrast, the policies calculated by our approach can provide decision support for different states and parameter developments, allowing companies to react to various scenarios. In the following, the state, the action, the cost function, and the information and transition function of the sequential stochastic problem are defined.

3.2.1. State

Based on the above considerations, we define the sequential stochastic optimization problem in detail. In each period t , the fleet $F_{tevy m}$ consists of a portfolio of vehicles. The vehicles are specified by their technology e , cluster v , age y , and mileage m . We denote the state S_t of the fleet as:

$$S_t = [F_{tevy m}]_{e \in \mathcal{E}, v \in \mathcal{V}, y \in \mathcal{Y}, m \in \mathcal{M}}. \quad (2)$$

S_0 shall be the fleet's initial state in period $t = 0$. The proposed algorithm leverages the so-called post-decision state. It describes the fleet's state after an action a has been decided and carried out. We denote it as S_t^a and slightly adjust the above equation:

$$S_t^a = [F_{tevy m}^a]_{e \in \mathcal{E}, v \in \mathcal{V}, y \in \mathcal{Y}, m \in \mathcal{M}}. \quad (3)$$

3.2.2. Action

In each period after the initial period $t = 0$, the company performs an action a_t regarding the number of purchases of new vehicles x_{tev}^p and the number of sales of old vehicles x_{tev}^s , subject to certain state-dependent constraints. Purchases and sales are modeled as integer variables. We define the action vector as follows:

$$a_t = [x_{tev}^p, x_{tev}^s]_{e \in \mathcal{E}, v \in \mathcal{V}}. \quad (4)$$

Subject to:

$$\sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} e_{tev}^o(S_t^a) \leq e^{og} e^{ob}, \quad \forall t = t^{gs}, \dots, t^{ge} \quad (5)$$

$$\sum_{v \in \mathcal{V}} d_{tv}^d \leq \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \sum_{m \in \mathcal{M}} l_{tev}^d F_{tevy m}^a, \quad t \in \mathcal{T} \quad (6)$$

$$\mathbb{E} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} (\hat{p}_{te}^p x_{tev}^p (1 - f_{te})) \leq c_t^{pb}, \quad t \in \mathcal{T} \quad (7)$$

$$x_{tev}^p = x_{tevy m}^p, \quad y = 0, m = 0, t \in \mathcal{T}, e \in \mathcal{E}, v \in \mathcal{V} \quad (8)$$

$$F_{tevy m}^a = 0, \quad y \geq Y, t \in \mathcal{T}, e \in \mathcal{E}, v \in \mathcal{V}, m \in \mathcal{M} \quad (9)$$

$$F_{tevy m}^a = 0, \quad m \geq M, t \in \mathcal{T}, e \in \mathcal{E}, v \in \mathcal{V}, y \in \mathcal{Y} \quad (10)$$

$$0 \leq x_{tev}^p \leq x_{tev}^{pmax}, \quad t \in \mathcal{T}, e \in \mathcal{E}, v \in \mathcal{V} \quad (11)$$

$$0 \leq x_{tev}^s \leq \sum_{y \in \mathcal{Y}} \sum_{m \in \mathcal{M}} F_{tevy m}, \quad t \in \mathcal{T}, e \in \mathcal{E}, v \in \mathcal{V} \quad (12)$$

$$x_{tev}^p = x_{tev}^s = 0, \quad t = 0, e \in \mathcal{E}, v \in \mathcal{V}. \quad (13)$$

Eq. (5) sets a goal for the fleet's operations emissions e_{tevy}^o . It is implemented as a percentage decrease e^{o8} in relation to a defined emissions baseline e^{ob} that must be met in years t^{8s} to t^{8e} . Naturally, multiples of these constraints can be applied and adjusted to company-specific circumstances. Eq. (6) ensures that the transport fleet can satisfy the demand d_{tv}^d (i.e., required distance) by cluster at each point in time. The constraint checks the demand against the maximum distance l_{tevy}^d a vehicle can travel in a period, based on its characteristics and operational circumstances. Eq. (7) implements a purchase budget allowing a maximum spend of c_t^{pb} on purchases each period. We assume that purchases include only new vehicles instead of used ones (i.e., age y and mileage m are zero), as can be seen in Eq. (8). This is in line with the standard operating procedure of the case study company. While the fleet ages, the constraints also ensure that no vehicle above a specific age Y and mileage M is retained (see Eqs. (9) and (10)). If the algorithm decides to sell more than these vehicles in a period, we assume that vehicles with the highest mileage in F_{tevyym} are sold first. These considerations are the reason why x_{tev}^p and x_{tev}^s are independent of age and mileage in our formulation. Finally, Eqs. (11) to (13) are the decision variables' domain constraints. The equations are used to limit the solution space and the computation time. While a negative number of purchases x_{tev}^p is naturally prohibited, we set an upper boundary x_{tev}^{pmax} based on an extrapolation of demand, including a reasonable buffer (Eq. (11)). The number of sold vehicles x_{tev}^s is bound to be non-negative, too, and may not exceed the number of vehicles in the (pre-decision) fleet (Eq. (12)). Note that these two constraints can be tailored to a specific case to decrease computation time further. For example, depending on the studied technologies and the existence and timing of zero-emissions targets, the purchases of polluting technologies can be set to zero. Similarly, the sales of polluting technologies need to be equal to the remaining vehicles of this technology in the fleet at times when zero-emissions constraints become effective. As we outline a generic model at this point, we refrain from specifying these additional constraints. Lastly, we set purchases and sales to zero for the initial period $t = 0$ (Eq. (13)). For simplicity, we exclude additional functional dependencies (i.e., apart from the dependence on t , e , v , y , and m) from the mathematical notation. Instead, we provide an overview of them in [Appendix B](#).

3.2.3. Costs

In a given period t , a fleet's TCO is a function of its state and the action taken (i.e., $C(S_t, a_t)$). We denote stochastic cost components with a hat and calculate the TCO based on the following generic equation:

$$C(S_t, a_t) = \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \sum_{m \in \mathcal{M}} (\hat{c}_{tev}^p - r_{tev}^f + \hat{c}_{tm}^b + \hat{c}_{te}^i + c_t^f + \hat{c}_{tev}^e + c_{tev}^m + c_{tev}^t + c_{tev}^d + c_{tev}^s + \hat{c}_{tevy}^r - \hat{r}_{tevy}^s + \hat{c}_{tevy}^c), \quad t \in \mathcal{T}. \quad (14)$$

To catch the subtle differences between alternative technologies, the TCO is composed of an exhaustive selection of cost (and revenue) types. It can be split into three categories: costs relating to purchasing activities, costs relating to operations, and costs relating to sales. Costs relating to purchasing include the actual vehicle purchase costs $\hat{c}_{tev}^p$, receivable funding r_{tev}^f , electric battery purchase costs $\hat{c}_{tm}^b$, infrastructure costs $\hat{c}_{te}^i$, and purchase fixed costs c_t^f . Operations costs include the costs for consuming fuel or energy $\hat{c}_{tev}^e$, vehicle maintenance costs c_{tev}^m , toll costs c_{tev}^t , driver costs c_{tev}^d , tax costs c_{tev}^s , and depreciation c_{tevy}^r . Lastly, we include the revenue from selling a vehicle $\hat{r}_{tevy}^s$. To complement traditional TCO categories, we integrate the cost of CO₂e emissions from all transport fleet activities in emission scopes one to three ($\hat{c}_{tevy}^c$). This includes emissions from vehicle production and sourcing activities, fuel production and sourcing, vehicle operations, and vehicle salvaging. To properly define the Markov decision process, we need to specify the cost calculation in more detail:

$$C(S_t, a_t) = \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \sum_{m \in \mathcal{M}} [(\hat{p}_{te}^p(1 - f_{te}) + e_{te}^p \hat{p}_t^{cn})x_{tev}^p + \hat{p}_t^b q_{tm}^b + \hat{c}_{te}^i + c_t^f + (k_{te} \hat{p}_{te}^e + c_{te}^m + c_{te}^t) d_{tev}^d + (c_t^d + c_{te}^s + \hat{c}_{tevy}^r) F_{tevyym}^a + e_{tevy}^o \hat{p}_t^{cn} + e_{te}^s \hat{p}_t^{cn} x_{tev}^s - \hat{p}_{tevy}^s x_{tevy}^s], \quad t \in \mathcal{T}. \quad (15)$$

When a vehicle of a specific technology is bought for a particular cluster (x_{tev}^p), we consider the stochastic purchase price $\hat{p}_{te}^p$ as well as available government funding f_{te} . The formulation accounts for the CO₂e emissions of all vehicle production and sourcing activities e_{te}^p , and monetarizes them with a stochastic carbon price $\hat{p}_t^{cn}$. If a number of electric batteries q_{tm}^b needs replacing, the cost of their purchase is calculated using the stochastic electric battery purchase price $\hat{p}_t^b$. In addition to these cost categories, the formulation includes fixed costs for purchasing c_t^f and infrastructure $\hat{c}_{te}^i$. Operations costs feature several cost types that depend on the distance driven, by cluster and technology d_{tev}^d . These include the cost of consuming a traditional or alternative fuel using the rate per distance k_{te} at the stochastic fuel price $\hat{p}_{te}^e$. Maintenance costs c_{te}^m and toll costs c_{te}^t are also multiplied by the distance driven. Driver costs c_{te}^d and tax costs c_{te}^s depend on the number of vehicles in operation in F_{tevyym}^a . The geometric degressive depreciation costs $\hat{c}_{tevy}^r$ also depend on the current fleet and consider the vehicles' age. Lastly, we calculate the CO₂e emissions of fleet operations e_{tevy}^o by multiplying it by the stochastic carbon price $\hat{p}_t^{cn}$. We account for the operation's well-to-wheel emissions, including the production and sourcing of all fuels, the vehicles' transport activities, and vehicle maintenance. Costs and revenues relating to sales activities depend on the number of vehicles sold x_{tev}^s . On the one hand, we include the CO₂e emissions of activities resulting from processes relating to salvaging e_{te}^s . On the other hand, we multiply the age-dependent sales price $\hat{p}_{tevy}^s$ of a vehicle of a particular technology with the number of sold vehicles x_{tevy}^s . See [Appendix B](#) for additional functional dependencies.

3.2.4. Information and transition function

Based on the cost equations, we can specify the vector of exogenous information W_t becoming known in period t . The vector comprises the stochastic cost components of Eq. (15). In addition, the EU ETS carbon price $\hat{p}_t^{ce}$ is included (see Appendix B for details):

$$W_t = [\hat{p}_{te}^p, \hat{p}_t^{cn}, \hat{p}_t^{ce}, \hat{p}_t^b, \hat{p}_{te}^e, \hat{c}_{tey}^e, \hat{p}_{tey}^s], \quad e \in \mathcal{E}, y \in \mathcal{Y}. \quad (16)$$

The modeling of these random parameters naturally depends on the respective case study. Hence, we revisit them in Section 4.2. We do not consider endogenous uncertainty in our model. Instead, we assume that all cost parameters are somewhat independent of decisions in earlier periods and that the parameters follow a defined trajectory. For the random parameters, this trajectory can still develop very differently in each iteration (based on the stochastic process definition), which could result from hypothetical other market players adopting AFVs in an earlier or later period. The modeled LSP would adjust its decisions based on these new circumstances (subject to the VFAs). There are many reasons for this choice of modeling. Above all, the road transportation sector is somewhat fragmented, and many small players run their own transport operations. Hence, cost parameters will not likely develop very differently when a single market player adopts AFVs earlier. Additionally, the case study company runs a small portfolio of trucks with few purchases and sales each period, so such an effect would be even smaller. Nevertheless, modeling endogenous uncertainty can be a worthwhile direction for future studies. It could be incorporated into our model by linking the number of purchases by technology with the parameters defining certain stochastic processes (e.g., BEV purchase price).

Next, we specify the transition function $S^M(S_t, a_t)$ of the sequential stochastic problem, describing how the system evolves from period t to period $t+1$. More precisely, it calculates the state of the next period S_{t+1} based on the state of the system S_t and the action a_t in the current period:

$$S_{t+1} = S^M(S_t, a_t), \quad \forall t = 0, \dots, T-1. \quad (17)$$

Subject to:

$$F_{(t+1)evym}^* = F_{tevm} + x_{tev}^p - x_{tev}^s, \quad (18)$$

$$F_{(t+1)evym} = F_{(t+1)ev(y+1)(m+d_{tev}^d)}^*. \quad (19)$$

The calculation is carried out in two steps. First, Eq. (18) calculates an interim fleet portfolio $F_{(t+1)evym}^*$. It features the fleet at the beginning of the current period F_{tevm} minus the vehicles sold x_{tev}^s plus the vehicles bought x_{tev}^p during this period. Afterward, this interim fleet $F_{(t+1)evym}^*$ changes its characteristics as it is used to run the transport operations throughout period t . The final fleet of the next period $F_{(t+1)evym}$ is achieved by increasing the vehicles' age y by one period and by increasing their mileage m by the distance driven d_{tev}^d (Eq. (19)).

3.3. Description of the approximate dynamic programming approach

The problem in Eq. (1) does not need to be solved in its entirety. Instead, in dynamic programming, any sequential stochastic problem can theoretically be solved to optimality by going backward through the time steps while recursively solving the so-called Bellman Equation:

$$V_t(S_t) = \min_{a_t} [C_t(S_t, a_t) + \gamma \mathbb{E}\{V_{t+1}(S_{t+1}) | S_t\}]. \quad (20)$$

In essence, it calculates the value V_t of being in a particular state S_t after an optimal decision a_t has been taken. This value is reached by optimizing the sum of the current period's contribution C_t and the value function V_{t+1} of the expected value of the next period over all possible actions a_t (Powell, 2011). However, the problem described in this paper suffers from the “curse of dimensionality.” With a growing number of periods, technologies, and clusters, it quickly becomes so large that it can no longer be solved using exact optimization methods (Powell, 2011).

Instead, we propose an ADP approach for solving the optimization problem. ADP builds on the Bellman Equation by using different approaches to replace the exact value function V_{t+1} with an approximation $\bar{V}_{t+1}$. This VFA is calculated iteratively, and learns to approximate the exact function more accurately with the number of model iterations (reinforcement learning) (Powell, 2011). We adopt a forward ADP approach relying on value iteration using the post-decision state variable suggested by Powell (2022). Algorithm 1 displays its pseudocode. The algorithm moves forward in time, continuously making decisions based on the iteratively updated VFA. Each period t is assigned a separate VFA, which learns independently of the other periods' VFAs. The model is run for N iterations. As a result of the applied approach, the notation introduced above is expanded. $\bar{V}_t^n(S_t^a)$ shall be defined as the value function approximation for period t in iteration n , which is updated using the step size α_n . The VFA is a function of the post-decision state S_t^a for period t after action a . We sometimes also use the notation $S_t^{a,n}$, if an indication of the iteration n is required. ω^n shall be different sample paths of the exogenous random information parameters (e.g., purchase prices) for iteration n which are defined by stochastic process models. The general exogenous information W_t is a function of these sample paths and the exogenous information for a particular iteration n can be written as $W_t^n = W_t(\omega^n)$. Lastly, the transition function $S^M(S_t^n, a_t)$ describes how the system transitions to an updated state, given a particular initial state S_t^n and a specific action a_t . To

Algorithm 1 Pseudocode of the applied ADP algorithm based on Powell (2022)

```

Initialize value function approximation  $\bar{V}_t^{n=0}(S_t^a)$  for all post-decision states  $S_t^a, t = \{0, 1, \dots, T\}$ 
Initialize initial state  $S_{t=0}^{a,n=1}$  for first iteration  $n$ 
for  $n = 1, 2, \dots, N$  do
    Choose a sample path  $\omega^n$ 
    for  $t = 1, 2, \dots, T$  do
        Determine action using dual annealing metaheuristic optimizing
            
$$\hat{v}_t^n = \min_{a_t \in A_t} [C(S_t^n, a_t) + \bar{V}_t^{n-1}(S^M(S_t^n, a_t))]$$

        Let  $a_t^n$  be the action that approximately solves the above minimization problem
        Update value function approximation subject to  $\bar{V}_{t-1}^n(S_{t-1}^{a,n}) = (1 - \alpha_{n-1})\bar{V}_{t-1}^{n-1}(S_{t-1}^{a,n}) + \alpha_{n-1}\hat{v}_t^n$ 
        Sample  $W_{t+1}^n = W_{t+1}(\omega^n)$  and compute next state  $S_{t+1}^n = S^M(S_t^n, a_t^n)$ 
    end for
end for
Return the value functions  $(\bar{V}_t^n)_{t=1}^T$ 

```

(approximately) minimize the TCO across all periods, the optimization problem boils down to iteratively optimizing the following equation for each period t and in each iteration n :

$$\hat{v}_t^n = \min_{a_t \in A_t} [C(S_t^n, a_t) + \bar{V}_t^{n-1}(S^M(S_t^n, a_t))], \quad (21)$$

where $\hat{v}_t^n$ is the calculated value obtained by finding the action a_t from the total action space A_t that minimizes the sum of current costs $C(S_t^n, a_t)$ and the value function approximation $\bar{V}_t^{n-1}(S^M(S_t^n, a_t))$, returning the estimated decision value in the period to come. As such, the model sequentially optimizes nested single-period lookahead problems until the end of the time frame T is reached. Note that the parameter developments are only revealed period by period, so the model does not know in advance which value a particular parameter will have in the subsequent periods. This procedure is then repeated for N iterations with different sample paths ω^n for the stochastic information. Note also that we adopt the notational style of modeling time proposed by Powell (2022), where physical processes evolve in discrete time steps, and information evolves continuously. For example, at $t = 1$, we can describe the discrete state of the physical system (denoted with the time index $t = 1$) and know the latest information that arrived continuously between $t = 0$ and $t = 1$ (which is also denoted with the index $t = 1$).

In contrast to Powell (2022), we apply a metaheuristic to choose a near-optimal action to minimize the sum of current costs and the VFA for the next period. In the next sections, we define the applied value function approximation and the metaheuristic.

3.3.1. Value function approximation

There are numerous approaches to designing VFAs which continuously learn to approximate the future value of a state (Powell, 2022). We decided to leverage VFAs based on a linear function, for several reasons. First, this approach is much simpler to implement and interpret than many more complex methods that often resemble a black box (e.g., neural networks). This argument is especially relevant considering the size and complexity of the problem, which may benefit from somewhat controllable and fast-to-compute VFAs. Second, the potential problem size also rules out implementation in the form of a simple lookup table. For each period, a separate table would need to be defined that efficiently clusters the huge solution space, requiring the storage of a vast and impractical amount of data. Third, a linear function can easily be tailored to the metric that is ultimately optimized, which is the TCO of the fleet. In many other ADP applications, the target value is more abstract or combines different metrics, so more complex VFAs may be beneficial.

The applied VFAs constitute the sum of k basis functions multiplied by the weightings θ_{tk}^{n-1} . As described, each period t is assigned an independent linear VFA, and each period's VFA possesses its own weightings. The weightings are updated once in each iteration n . We rely on a decomposition of the transport fleet's TCO into $k = 13$ different cost types, which we have already introduced in Eq. (14). The resulting equation for the VFAs is given by:

$$\begin{aligned} \bar{V}_t^{n-1}(S^M(S_t^n, a_t)) = & \theta_{t1}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \hat{c}_{tev}^p - \theta_{t2}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} r_{tev}^f + \theta_{t3}^{n-1} \sum_{m \in \mathcal{M}} \hat{c}_{tm}^b + \theta_{t4}^{n-1} \sum_{e \in \mathcal{E}} c_{te}^i + \theta_{t5}^{n-1} c_t^f + \\ & \theta_{t6}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \hat{c}_{tev}^e + \theta_{t7}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} c_{tev}^m + \theta_{t8}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} c_{tev}^t + \theta_{t9}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} c_{tev}^d + \\ & \theta_{t10}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} c_{tev}^s + \theta_{t11}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \hat{c}_{tevy}^r - \theta_{t12}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \hat{r}_{tevy}^s + \\ & \theta_{t13}^{n-1} \sum_{e \in \mathcal{E}} \sum_{v \in \mathcal{V}} \sum_{y \in \mathcal{Y}} \hat{c}_{tevy}^c. \end{aligned} \quad (22)$$

While simulating the periods and iterations, the weightings of each period are continuously updated (once each iteration) using a recursive least squares method based on the Sherrman–Morrison formula proposed by Powell (2022). We outline the main equations

below, but refer to the original publication for details. θ_t^n shall be the k -dimensional vector of weightings from the VFA (Eq. (22)) for a single period t . Its iterative update follows the equation:

$$\theta_t^n = \theta_t^{n-1} - H_t^n \Phi_t^n \hat{\epsilon}_t^n, \quad (23)$$

where Φ_t^n is the vector of the k basis functions (see coefficients in Eq. (22)) and $\hat{\epsilon}_t^n$ is the stochastic error between forecast from period $t-1$ and actual values in t . H_t^n is a matrix recursively calculated by:

$$H_t^n = \frac{1}{\gamma_t^n} J_t^{n-1}. \quad (24)$$

γ_t^n comprises a scalar variable following the equation:

$$\gamma_t^n = 1 + (\phi_t^n)^T J_t^{n-1} \phi_t^n, \quad (25)$$

and J_t^n is another $k \times k$ matrix updated recursively following the equation:

$$J_t^n = J_t^{n-1} - \frac{1}{\gamma_t^n} (J_t^{n-1} \Phi_t^n (\Phi_t^n)^T J_t^{n-1}). \quad (26)$$

In our implementation, the weights are initialized with a value of 1.0 in the first model iteration. Matrix J_t^0 is initialized as the identity matrix multiplied by a tunable scalar. We also introduce upper and lower boundaries for the values of the weightings. These limits represent additional hyperparameters to be tuned. We allow weightings between zero and 20. In our numerical calculations, this measure ensures that the components keep their correct algebraic sign, and prevents overfitting.

3.3.2. Metaheuristic for iterative optimization

In a standard ADP implementation, the iterative optimization in each period is carried out by (backward) dynamic programming, which is an exact optimization algorithm. In addition, the ϵ -greedy strategy is applied. Within ϵ -greedy, there is a certain probability ϵ that the algorithm picks a random solution instead of following the dynamic programming solution (Powell, 2022). The goal is that the algorithm sufficiently explores the solution space. Otherwise, it might only pick the best solution in every single period but fail to visit and eventually choose a solution that is (near-)optimal across all periods. Again, the potential size of the fleet renewal problem with AFVs prohibits us from choosing this standard approach. On the one hand, dynamic programming finds the exact solution for each period, but it takes too much time to do so. It cannot efficiently be used to iteratively solve the problems in each time step of the ADP algorithm. On the other hand, by simply choosing a random action, ϵ -greedy only rarely produces valuable decisions for large solution spaces. A simple example illustrates this. Assume we want to optimize the renewal of a fleet comprising a single cluster over a number of periods. Each period, we decide which vehicles to sell and which vehicles to purchase. Following our problem formulation, sales (by technology) may range between zero and the number of vehicles in the fleet of technology e . Purchases may range between zero and (slightly above) the total number of fleet vehicles. Assume that ϵ -greedy only proposes decision vectors satisfying the problem constraints (e.g., regarding sales of too old vehicles). Assume we are in period t and the constraints demand that ten vehicles of some technology need to be sold because of age or mileage constraints. In total, the fleet shall include 100 vehicles of technology e . First, ϵ -greedy will propose a random number of sales x_{te}^s between ten and 100. The probability that it will choose exactly ten is obviously small. However, the probability that it will choose a number, for example, above 30, is very high. Each sale above ten means a vehicle is sold that has not reached its age or mileage constraints and could have been used for more periods. Moreover, the higher the number of sales, the younger the vehicles sold (because the oldest are sold first). This can make sense if the financial benefits of a particular new technology are suddenly much higher than those of the currently used technologies. However, this is unlikely as we continuously make decisions each period (or year). On top of that, assume that technology e is actually the one that will be the most beneficial in the future. The probability that we will keep it in the fleet and not sell it (i.e., $x_{te}^s = 0$) or at least only sell a few vehicles is very small. Additionally, by frequently selling a high number of vehicles, we also need to purchase many new vehicles to satisfy demand. As a result, the total number of purchases in the timeframe increases. Because the purchase costs comprise by far the most influential TCO category, the TCO skyrockets. This effect is amplified in the next step. After determining the total number of sales across technologies, ϵ -greedy tries purchasing a number of new vehicles x_{te}^p to fulfill the total demand. For each technology, ϵ -greedy will pick a random number of purchases x_{te}^p between zero and (slightly above) the total fleet size. Again, the probability that we purchase a number of vehicles similar to the number that has been sold is minimal. Instead, we will likely end up with more vehicles than we actually need to satisfy the demand. This naturally induces unnecessary costs, making the fleet inefficient. Like before, each technology has the same chance to be purchased in the random algorithm. Of course, there might be technologies that are close to each other regarding their total cost position. However, this will likely not be the case for all technologies. So, we will most certainly end up with many vehicles of inferior technology. In total, these considerations show that ϵ -greedy cannot produce helpful decision vectors in our study.

Instead of the standard strategy, we leverage a metaheuristic to perform the optimization within single periods. Compared to normal dynamic programming, the problems become computationally tractable. However, the calculated solutions are (most certainly) only approximate to the optimal solution. Compared to ϵ -greedy, the metaheuristic guides us close to the optimal decisions regarding the ideal number of purchases and sales by technology. Because the value function steers the metaheuristic, it more likely prefers beneficial technologies. However, it still maintains opportunities to explore new technologies and higher or lower numbers of purchases and sales (as it yields only an approximate solution).

We apply a dual annealing metaheuristic described by Xiang et al. (2000, 2013). The program code is publicly available via GitHub (Gubian and Xiang, 2018). Dual annealing is a modification of the well-known simulated annealing, which artificially recreates the annealing process in metallurgy or material science (Xiang et al., 1997). The traditional simulated annealing algorithm starts with a high temperature c , which is slowly decreased. It is used as a probability for the acceptance of sub-optimal interim solutions to the optimization problem (Gendreau et al., 2019). Dual annealing combines a local search approach with the so-called generalized simulated annealing, which itself is a combination of two other simulated annealing types, namely classical simulated annealing and fast simulated annealing. The generalized simulated annealing generalizes the two types by leveraging the Tsallis statistics and adopts a deformed Cauchy–Lorentz visiting distribution which controls how the algorithm searches through the solution space:

$$g_{q_v}(\Delta x(t)) \propto \frac{[T_{q_v}(t)]^{-\frac{D}{3-q_v}}}{\left[1 + (q_v - 1) \frac{(\Delta x(t))^2}{[T_{q_v}(t)]^{\frac{2}{3-q_v}}}\right]^{\frac{1}{q_v-1} + \frac{D-1}{2}}}, \quad (27)$$

where the parameter q_v controls the shape of the visiting distribution. The temperature T , the solution space dimensions D , and the trial jump distance Δx predominantly determine which solutions will be tried out. The temperature, or rather its decrease, is defined by the equation:

$$T_{q_v}(t) = T_{q_v}(1) \frac{2^{q_v-1} - 1}{(1+t)^{q_v-1} - 1}. \quad (28)$$

Finally, the probability p_{q_a} to accept a (suboptimal) solution requiring the uphill climbing energy ΔE relative to its difference to the current optimum is given by:

$$p_{q_a} = \min\{1, [1 - (1 - q_a)\beta\Delta E]^{\frac{1}{1-q_a}}\}. \quad (29)$$

It depends on the tunable acceptance parameter q_a (Xiang et al., 2013). The solutions from the generalized simulated annealing are afterward fed into a local search procedure to get as close as possible to the global optimum (Xiang and Gong, 2000; Xiang et al., 2013).

4. Case study description

The generic theoretical model outlined in the previous sections is applied to a real-world case study. The primary goal is to test its validity. If the model produces reliable and meaningful results, the case study can be used to derive guidance for companies and policymakers. The case study concerns an anonymous German LSP with an internal target to reduce CO₂e emissions from its truck fleet over the coming years. From 2030, the company wants to operate with 65% of its 2019 emissions baseline. From 2045, it aims to run emissions-free. To reach these targets, the company's fleet managers must develop strategies to integrate AFVs into the regular decisions on which trucks to sell and which new trucks to purchase. To include the target years and sufficient periods thereafter, we choose to model the fleet's mobility transition on a timeline until 2050. The fleet is of significant size, with well over 200 trucks. Currently, almost all trucks are diesel-powered, and only 2% of the fleet is powered by an alternative fuel. BEVs have recently been integrated into the fleet as part of a pilot project. For the optimization, we consider four competing technologies: ICEV, LNGV, BEV, and FCEV. This selection reflects company preferences, and encompasses the most promising alternatives in the distant future (i.e., in 20 or more years), in line with various previous studies (Anderhofstadt and Spinler, 2019; Peters et al., 2021; Bae et al., 2022). The company's truck fleet consists purely of heavy-duty trucks with a total permissible weight above 20 tons. Multiple company locations, mainly in Germany, act as strictly assigned truck hubs. The vehicles are deployed to serve customers in the regional vicinity of the hubs, on closed-loop tours that consist primarily of multiple stops. For example, the trucks supply customer production plants with raw materials, and transport goods from (inland) harbors to customer locations. The company plans to build the required infrastructure for BEVs and FCEVs on its own and at its hub locations. That also means that, for example, BEVs can be charged overnight. Owing to the strict hub assignment, the regional nature of clients, and the absence of strong season- or trend-related effects, the characteristics of each truck's tours are very stable throughout the year (e.g., distance traveled, loaded weight, number of stops). Nevertheless, these characteristics vary considerably between trucks. The average annual distance traveled by trucks in the fleet ranges from 20,000 to 170,000 km, with an overall average of 80,000 km. Similarly, the average loaded weight per tour varies between 15 and 38 tons, with an overall average of 26 tons.

The circumstances of the decarbonization in the case study are very common in the heavy-duty trucking segment. In particular, similar timelines and emissions targets can be observed for several leading German LSPs. For example, DB Schenker aims to be carbon-neutral by 2040. They set an interim target for road transportation of 50 percent by 2030 (The Climate Choice UG, 2022). Deutsche Post DHL Group wants to achieve net-zero emissions by 2050. By 2030, 30 percent of the consumed fuel shall come from sustainable sources (Deutsche Post DHL Group, 2022). Other countries operate on similar timelines. In the US, the most ambitious target for decarbonization is present in California. The Advanced Clean Fleets rule is supposed to electrify trucking fleets by 2042, which is considered an enormous challenge (Canary Media, 2023). It becomes clear that the trucking segment is far more complicated to decarbonize than the last-mile delivery segment, where companies like Amazon or DHL already operate large sustainable fleets economically. Our research on the case study can contribute to closing this gap by providing valuable guidance for effective decarbonization policies in the heavy-duty trucking segment and other areas that are more difficult to run with AFVs.

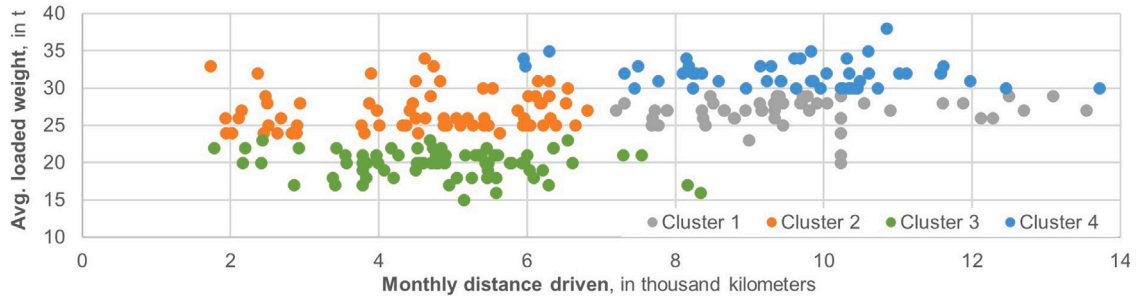


Fig. 2. Results of the clustering analysis.

4.1. Defining the operational vehicle clusters

We define vehicle clusters to capture the influence of varying operational circumstances following the generic problem definition (see Section 3.2). We derive the clusters by leveraging hierarchical and k-means clustering. We use the average distance driven in a representative month, and the average loaded weight per tour as features, as suggested by a prior correlation analysis. The results of the clustering are shown in Fig. 2. The applied methods propose four clusters, effectively dividing the total fleet into four quadrants. Cluster 3 has the lowest average load and a relatively low average distance. In cluster 2, the trucks drive similar distances, but their loads are considerably heavier. In comparison, the distances driven in clusters one and four are much longer. Whereas cluster 1 transports average loads similar to those in cluster 2, the trucks in cluster 4, on average, carry much heavier shipments than the others.

In the case study, the clusters are optimized sequentially to save computation time, which would otherwise increase significantly due to a much larger combined solution space. Consequently, we apply a couple of additional mechanisms to ensure that the company emissions goal is achieved and we stay as close as possible to an optimal solution. First, the clusters are naturally linked by the aggregate emissions constraint. Secondly, each cluster can emit a particular proportion of the company's aggregate emissions target, which is equal to the distance driven within the cluster, divided by the distance driven across all clusters. However, the algorithm can violate these individual emissions targets if the aggregate emissions target of the whole fleet is satisfied. Thirdly, both constraints are implemented as penalties in the target function, while the aggregate constraint has a bigger assigned penalty. That means the constraints can be violated but only at a significant cost. Fourth, we force the last cluster in the optimization sequence to try to satisfy both constraints (individual and aggregate). This ensures that the aggregate constraint is met, as far as it is in the power of the last cluster. Fifth, we randomly choose a sequence to optimize the clusters. We do this to avoid that one particular cluster is always decarbonized first only because it is at the end of the sequence and needs to make up for the underperforming previous clusters. Lastly, in combination with the previous points, we benefit from the intelligent VFAs. They learn that they should try to achieve both the individual and the aggregate constraints to avoid heavy penalties. They also learn which clusters to decarbonize first in order to minimize costs.

4.2. Modeling the stochastic processes

Naturally, the program must be tailored to the specifics of the case study. For example, depending on the size of the deployed trucks, different cost values and developments must be specified. Rather than explaining the many parameters in detail at this point, we explain the methodology for modeling only the stochastic parameters. An overview of the sources used to inform all of the parameters is provided in the online appendix. In total, we model 11 independent stochastic processes: the vehicle purchase price by technology (ICEV, LNGV, BEV, FCEV), fuel prices by energy source (diesel, liquified natural gas [LNG], electricity, hydrogen), the purchase price for electric batteries, and two carbon prices (EU emissions trading scheme [EU ETS], German national emissions trading scheme [nEHS]). The EU ETS influences the electricity price in the model. The nEHS price influences the price of other fuels and is used as a carbon price in Eq. (15). Note that we exclude the depreciation costs $\hat{c}_{ley}^r$ and the sales price $\hat{p}_{ley}^s$ from this list. Both are not modeled with an independent stochastic process but instead largely depend on the stochastic process of the purchase price $\hat{p}_{ley}^p$.

We model all 11 parameters as separate Ornstein–Uhlenbeck processes. Ornstein–Uhlenbeck processes have been used in similar contexts many times and, as such, represent proven and valid instruments for modeling technological transitions. Ansariipoor and Oliveira (2018) apply an Ornstein–Uhlenbeck process to model BEV lease costs. They argue that over a long time horizon the cost of batteries and the leasing costs of BEVs decrease at a certain speed and, at some point in time, reach a stable level. Similar developments have been observed for the costs of other technologies, such as ICEV costs. Kleindorfer et al. (2012) and Neboian and Spinler (2015) use an Ornstein–Uhlenbeck process to model battery prices in a fleet renewal setting to capture technological progress. Geursen et al. (2023) use it to model demand and fuel prices in an airline fleet planning setting. Falbo et al. (2021) model the total opportunity cost for adopting electric vehicles with such a process. Ketterer (2014) applies it to forecast electricity prices and their expected development depending on increased wind power generation in Germany.

Irrespective of past publications, it is clear that the 11 parameters are all challenging to forecast up to 2050. Their development depends heavily on many uncertain factors, such as technology readiness, market penetration, and the type and severity of regulations. Currently, there are no legal targets for heavy-duty truck fleet managers in Germany. The emissions targets we apply are company-specific and would further be individual from company to company. The companies would thereby not execute their mobility transitions all at once, but we would rather see diverse timings and degrees. As a result, we do not expect discontinuous price jumps or technology shortages in the time horizon. We rather expect a relatively smooth transition driven by the vast portfolio of company ambitions and technological progress. While the alternative technologies start to mature (i.e., technological capability, infrastructure readiness, etc.) and the market slowly adopts them, prices are declining at a particular pace. At some point, full maturity is reached, and the market regularly uses alternative technologies. At this time, prices will converge to a certain average level. That does not mean that they do not fluctuate. There is still an up and down introduced by the noise term of the stochastic process. However, assuming there are no legal changes, political crises, or severe inflation, a steady average is achieved.

Mean-reverting Ornstein–Uhlenbeck processes are ideal for modeling the described behavior and creating parameter developments tailored to the characteristics of the 11 parameters. Consequently, we model the processes based on the following stochastic differential equation based on the work of Vasicek (1977):

$$dZ_t = \kappa(\mu - Z_t)dt + \sigma I_t. \quad (30)$$

In this equation, Z_t shall be the modeled parameter value (e.g., BEV purchase price) in period t . κ is the mean reversion rate defining how quickly the process reaches the final value. For example, a larger κ could be used to model relatively low BEV purchase prices achieved in the early periods of the time horizon (e.g., caused by faster acceptance or faster technology development). μ is the mean reversion level, representing the value around which the process evolves and will eventually reach. Diffusion σ characterizes the randomness or volatility of the process, and I_t is a standard Wiener process (Vasicek, 1977; Weron, 2014; Ketterer, 2014).

For each stochastic parameter, information on current and future potential values and developments is gathered from company data, expert inputs, and existing scientific literature. The data is triangulated to form corridors in which the stochastic parameters may take on different random developments up to the final period of 2050. Note that the corridors evolve over time. Their minimum and maximum values depend on the time periods and cater to the specific circumstances of the individual parameters. Thereby, the corridors already account for the influence of factors like technological development, the creation of infrastructure, or the rising demand for green alternatives in society. In addition to that, we model quite wide corridors, as is appropriate for the considered time horizon. The Ornstein–Uhlenbeck processes are then fitted to these corridors by adjusting the arguments of Eq. (30) accordingly. The supplementary material provides a detailed description of the modeling of each parameter, including inputs, assumptions, and sample developments. Again, we would like to highlight the difficulty of forecasting these parameters and remind that we intend the model to be used in an iterative manner (e.g., each year, every second year, etc.). Its assumptions and forecasts (incl. stochastic processes) need to be updated continuously to define and adjust the fleet renewal strategy based on the latest and most accurate information.

5. Computational experiments

This section describes the setup and results of the computational experiments conducted in the case study context. First, we prove the validity of our approach in comparison with benchmark approaches for a smaller problem setup. We then describe the results for the holistic problem. We conclude the section with the outcomes of sensitivity analyses run for selected model parameters. All models are coded in Python and all experiments are executed on a Windows workstation (Intel Core i5 1.6 GHz CPU and 16 GB RAM).

5.1. Validation of the approach on a reduced problem

The large size of the holistic problem makes it challenging to prove the validity of the approach. A typical procedure compares the convergence and quality of results with alternative methods such as different exact algorithms. As indicated, in this case the problem is computationally intractable for exact methods, so such a comparison is impossible. To overcome this issue, we define a smaller problem by reducing its dimensions. We then apply the proposed ADP model to the reduced problem and compare its performance with multiple benchmark approaches.

The reduced problem features one cluster (cluster 3), two technologies (ICEV and BEV), 29 time periods (2021 to 2050), and a deterministic rather than stochastic setting. We solve it with five methodological approaches: the proposed ADP method, an exact greedy method, and three heuristics. The ADP model is run for 100 iterations, and its convergence is tracked. On average, a single iteration takes 5.4 min to compute, so 100 iterations already take 9.1 h for the reduced problem. The exact greedy method always picks the fleet portfolio that is cost-optimal for the simulation period, optimizing immediate rather than future rewards. For reference, running this method takes 7.0 days. The three heuristics follow simple basic principles. In each period, they sell what needs to be sold, for example when vehicles pass the maximum age or mileage or violate the emissions constraint. They also try to hold vehicles for as long as possible, and they only buy the minimum number of trucks needed to fulfill the demand. The three heuristics differ in terms of their purchasing strategies. The first one only buys ICEVs until 2030, and purchases exclusively BEVs thereafter (Heuristic 1). The second buys BEVs and ICEVs equally until 2030, and only BEVs thereafter (Heuristic 2). The last relies purely on BEVs (Heuristic 3). Naturally, the three heuristic solutions are computed in a matter of seconds.

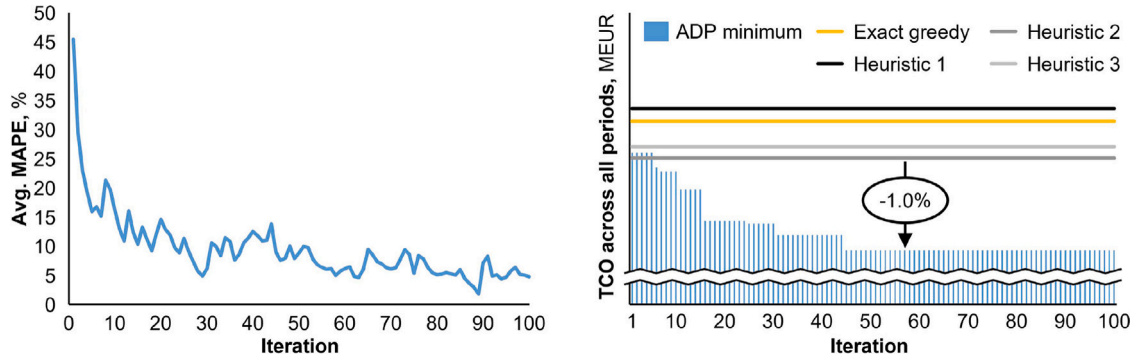


Fig. 3. MAPE development of VFAs (left) and TCO of compared methods for the reduced problem (right).

The lefthand side of Fig. 3 displays the average development of the ADP value functions' mean absolute percentage error (MAPE) throughout 100 iterations. The MAPE measures the error between a VFA's forecast in period t and the actual value in period $t + 1$. The MAPE starts at 45% and quickly declines to a forecast error below 10% in iteration 18. By iteration 55, it eventually converges to around 6%. This development illustrates that the VFAs can learn the future TCO of the fleet and quickly produce high-quality forecasts with small errors. In addition, the results of the benchmarked methods are presented on the righthand side of Fig. 3. The chart plots their TCO along the iterations. Complementing the MAPE development, the ADP continuously learns to find better solutions until it reaches a minimum around iteration 45. This minimum dominates all solutions calculated by the benchmark methods, and beats the next best solution produced by the second heuristic by roughly 1.0%. Interestingly, the ADP is surpassed only by the same heuristic in the first iterations, demonstrating that it already produces valuable solutions without requiring much learning. Ultimately, the ADP finds the best timing and speed of BEV adoption among the alternatives compared. The exact greedy method provides the best solution for every single period. However, in doing so, it fails to find the perfect transition point towards BEVs, and thus the best policy across all periods. The heuristics surpass the exact greedy method by taking risks and applying BEVs earlier (heuristics 2 and 3).

In conclusion, these experiments validate the proposed ADP approach and indicate that it produces high-quality fleet renewal policies that can create tangible value for practitioners. It also unlocks the possibility of finding solutions to even bigger problems that take account of multiple technologies and clusters in a stochastic setting, as described in the next section.

5.2. Numerical model results for the holistic problem

We apply our approach to approximately solve the holistic problem as part of the LSP case study. We consider all stochastic parameters, four technologies, and four vehicle clusters over 29 time periods. The model is run for 100 iterations. Calculating a single iteration takes 11.6 h, so 100 iterations are calculated in 6.9 weeks. Again, this highlights how complex the problem is and why exact approaches cannot solve the problem in an acceptable time. As before, the MAPE of the periods' VFAs is constantly tracked, showing a steep decline in the first model iterations. The error is already below 10% in period six, revealing that good forecast quality is reached relatively early in the experiment. Later, the MAPE fluctuates predominantly between five and 10%, and converges to around 5%. In the following analyses, we exclude the first iterations to focus only on those with a sufficiently small error size. We discuss the average model outcomes across all relevant iterations. They can be regarded as a proposed policy assuming the average realization of all stochastic parameters. Remember that one of the approach's key strengths is providing decision support considering particular problem states and parameter developments based on the policy function (represented by the trained VFAs). As such, the below results only reflect one potential (average) outcome. In the online appendix, we include an analysis of extreme adoption rates, which demonstrates the model's guidance for alternative future scenarios.

Fig. 4 shows the average yearly adoption of the four technologies for all clusters and across all relevant iterations. It is measured as the number of trucks of a particular technology divided by the number of trucks across all technologies. The initial portfolio consists almost entirely of ICEVs. Until 2029, this technology remains dominant, but its share steadily decreases to 77%. At the same time, there are slight increases in LNGV and BEV adoption, suggesting that these technologies may make sense for selected cases. In 2029, 13% of the fleet consists of LNGVs and 11% of BEVs. FCEVs are not competitive in the early periods. When the first CO₂e constraint takes effect in 2030, many ICEVs must be sold off, and the ICEV adoption sinks to 37%. In contrast, BEV adoption rises significantly to 53%, making it the dominant technology. In the following phase, the proportions of BEVs and LNGVs in the fleet remain relatively constant, while the ICEV share slowly decreases. Note that the fleet constantly grows, resulting from increasing demand or distance to be covered each year. A stable share is thus equivalent to additional purchases, whereas a decreasing share may mean that the absolute number of vehicles is relatively stable. The deployment of FCEVs rises slowly. In 2035 it reaches 3%, and by 2039 it achieves a similar level to LNGVs (around 10%). 2040 is an important year in the modeled mobility transition, as FCEV adoption increases to 24%, while BEV adoption falls to 38%. FCEVs are evidently becoming more and more competitive. In

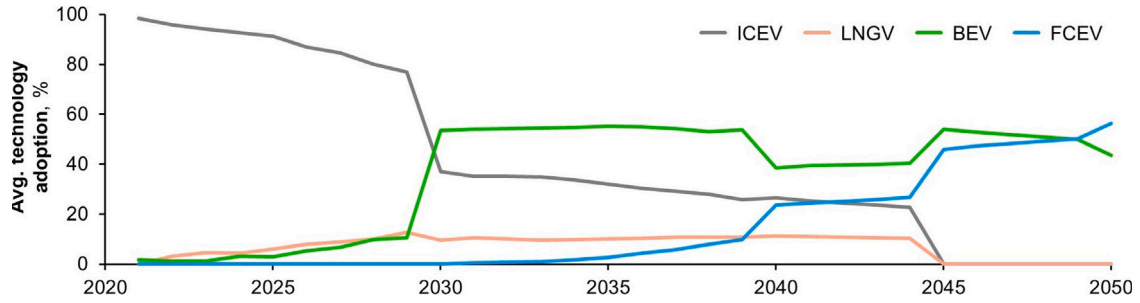


Fig. 4. Average technology adoption across clusters and iterations.

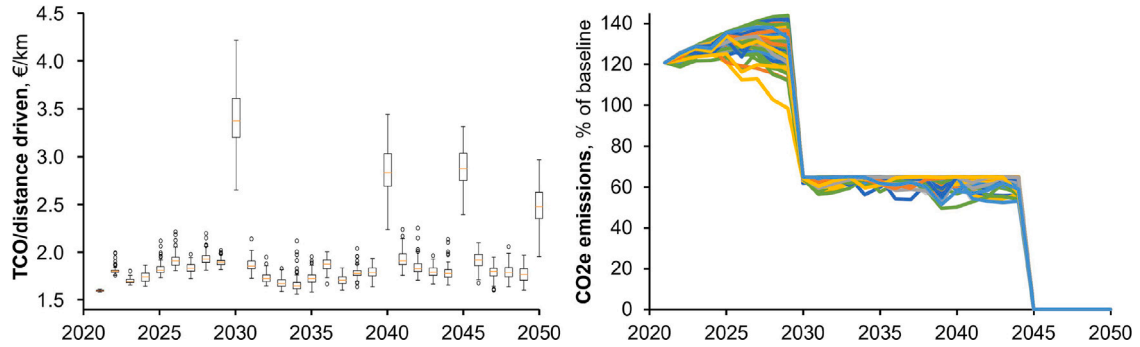


Fig. 5. Box plot of annual TCO per distance driven (left) and CO₂e emissions of all iterations relative to the emissions baseline (right).

2042, the fleet accommodates more FCEVs than ICEVs for the first time. When the zero-emissions constraint takes effect in 2045, all remaining ICEVs and LNGVs are sold off, again leading to a leap in FCEV and BEV adoption. Although BEVs remain the dominant technology at 54%, the gap to FCEVs (46%) gradually diminishes. In the remaining years, FCEV adoption constantly increases at the expense of BEV adoption. FCEVs become the dominant technology in the fleet in 2050, with an adoption of 56%.

The proposed fleet renewal policy has a significant impact on the company's cost position. On the lefthand side of Fig. 5, this impact is illustrated with a boxplot of the fleet's annual TCO per distance driven across all relevant iterations. We observe four peaks in the chart. The peaks in 2030 and 2045 are explained by the CO₂e emissions constraints and the fact that the majority of purchases required to satisfy these constraints are made at the last moment. This occurs for two reasons. First, the algorithm has learned that it is cheaper to hold vehicles for as long as possible, ideally for their entire lifetime (i.e., 10 years or one million kilometers in our assumptions). In doing so, it profits from fewer purchases and sinking prices. The second reason is revealed in the righthand side of Fig. 5, which shows the fleet's CO₂e emissions in all iterations. On average, the algorithm tries to operate the fleet as close as possible to the initial emissions constraint (i.e., at 65% CO₂e emissions). It effectively uses BEVs and FCEVs to fulfill the constraints, but still builds up ICEVs and LNGVs as much as possible to minimize costs. This decision reveals that combustion engine technologies still have a significant cost advantage, fostering high purchases in these two years. Similar reasons cause peaks in the TCO in 2040 and 2050. As the trucks are preferably held for their entire lifetime, many of the trucks bought in 2030 because of the first carbon constraint must be replaced in 2040. The same occurs ten years later, in 2050. The four TCO peaks already highlight that executing the mobility transition will require substantial investment. However, even if these are disregarded, the chart indicates a slight trend towards higher TCO per distance. In addition, the TCO per distance evolves in waves, caused by natural fleet aging and the corresponding number of mandatory purchases and sales.

Fig. 6 displays the TCO per distance that a truck using a particular technology could achieve in each single period. For the analysis, we average all stochastic cost parameters by period to better reflect average adoption. The values are shown as ranges explained by different purchase timings. When a truck has been bought recently, its TCO per distance is likely to appear at the upper end of the range because high fixed costs are spread across shorter distances driven. The range's lower end is defined by trucks that have operated for a long time or were purchased after substantial cost decreases. The chart effectively explains many previous observations. FCEVs remain uncompetitive for some time. High initial infrastructure costs and high vehicle purchase costs in the case study are the main reasons for this. Infrastructure costs also cause the extensive FCEV cost range. The more FCEVs are deployed over time, the more the investment can be spread across them. In 2045, the TCO per distance of a new FCEV (5.11 €/km) falls below that of a new BEV (5.16 €/km) for the first time. The main reason for this is carbon costs. Both technologies cause considerable emissions during truck production. However, BEV production evidently emits more CO₂e than FCEV production, caused mainly by battery manufacturing (Booto et al., 2021; Rial and Pérez, 2021). We infer that the formulation of Eqs. (15) and

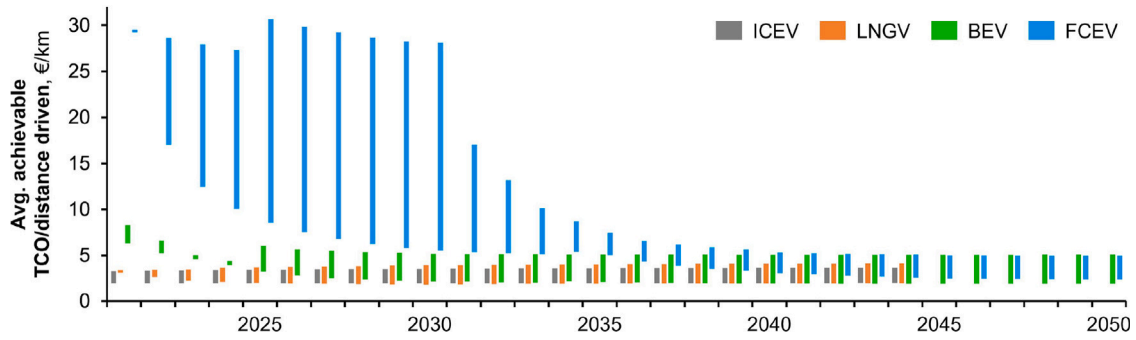


Fig. 6. Technology-specific achievable TCO per kilometer driven for an average cost parameter development.

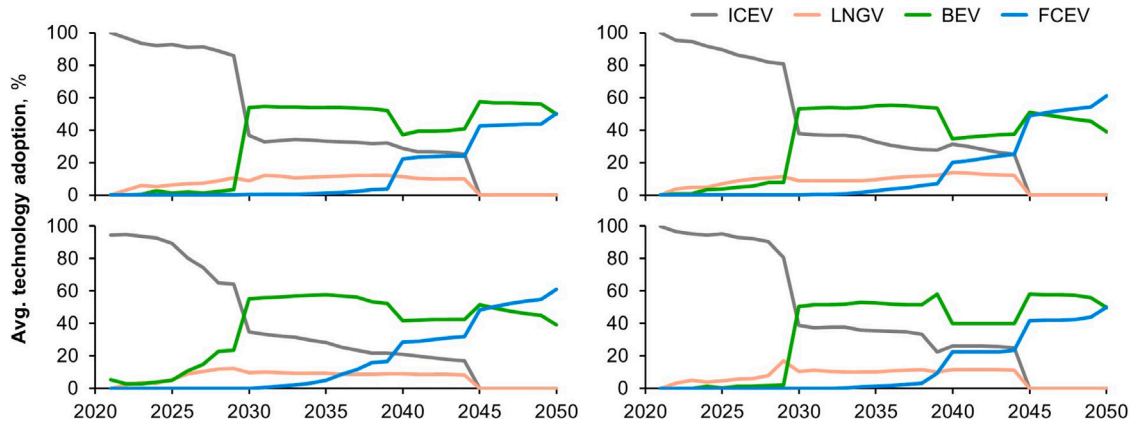


Fig. 7. Average technology adoption for vehicle clusters one (top left), two (top right), three (bottom left), and four (bottom right).

(22) benefits FCEVs compared with BEVs by accounting for end-to-end emissions. In 2050, a new FCEV ultimately costs 5.04 €/km, whereas a new BEV costs 5.21 €/km. However, BEVs beat all other technologies at the lower boundary of the cost range from 2040 onwards (1.75 €/km to 1.78 €/km). In the same timeframe, the minimum for FCEVs remains between 2.20 €/km and 2.94 €/km, for ICEVs between 1.83 €/km and 1.85 €/km, and for LNGVs between 1.79 €/km and 1.83 €/km. This comparison highlights that BEVs have an advantage in variable costs. Nevertheless, their high fixed costs (especially purchase and cradle-to-gate carbon costs) make them inferior until later periods of their deployment. ICEV and LNGV cost ranges increase slightly throughout the time horizon of their availability, but remain at comparatively low levels overall. Their cost ranges are much smaller because of their lower fixed costs (especially purchase, infrastructure, and cradle-to-gate carbon costs). A new LNGV always remains more expensive than a new ICEV. For example, in 2040, a new LNGV can be operated at 4.21 €/km and a new ICEV at 3.76 €/km. In addition, LNGVs have a slight variable cost advantage from 2026 onwards (mainly fuel and carbon costs), bringing the lower end of their cost range below that of ICEVs. Again, the algorithm prefers the technology with lower fixed costs. Finally, note that the upper limits of BEV and FCEV costs before 2025 are slightly lower than in the periods thereafter. Currently available funding explains this: the subventions are insufficient to close the cost gap, and fail to sustainably increase the adoption of AFVs.

Separate inspections of the four vehicle clusters are necessary to analyze whether and how their operational characteristics influence fleet renewal policies. Fig. 7 divides the average adoption values shown in Fig. 4 into the four clusters. At first sight, there are no immense variations in technology adoption. We conclude that operational differences between the clusters are not sufficiently significant to induce stark changes to fleet renewal policies. As outlined earlier, each cluster operates on comparatively short tours around specific hubs. The total distances covered annually are considerably smaller than those of typical medium- to long-range operations of other LSPs. Nevertheless, some interesting observations can be made. Cluster 3 decarbonizes slightly faster than the others. By 2029, it already reaches a fleet decarbonization of 23%. It also considers FCEVs much earlier, proved, for example, by an adoption of 17% in 2039. The cluster is characterized by a low monthly distance and low loaded weight. It also features the highest number of trucks. The algorithm has evidently learned to prioritize decarbonization of this cluster over the others, and uses it to reach the company's aggregate carbon constraints more easily. Furthermore, low-range clusters two and three show a slightly different adoption balance in the latter periods. Here, the algorithm more clearly prioritizes FCEVs over BEVs, which is explained by higher carbon costs in BEV production.

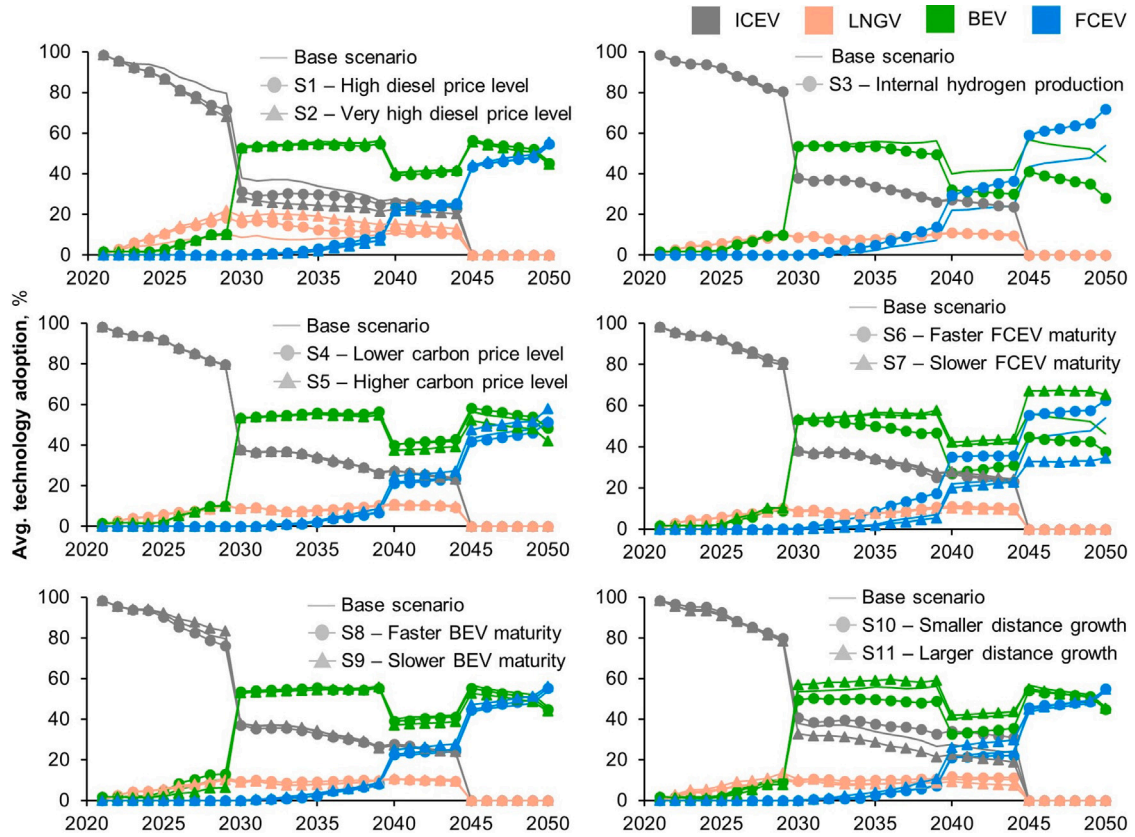


Fig. 8. Average technology adoption in the 11 sensitivity scenarios.

5.3. Sensitivity analysis for selected parameters

Complementary to the base model results, we conduct sensitivity analyses of selected parameters. Owing to the problem's large size and associated long program runtime, we limit these to a manageable number of 11 sensitivity scenarios identified in collaboration with the case study company. These represent alternative extreme scenarios based on the literature review, which was also used to define the parameters of the base model. Detailed descriptions of the sensitivity scenarios with the most significant effects are presented below. The online appendix gives details of the selection and modeling of all scenarios and the results for less significant sensitivities.

The scenarios' effects on technology adoption are shown in Fig. 8. The first two scenarios feature high (S1) and very high (S2) mean reversion levels of diesel prices. These two scenarios significantly influence only ICEV and LNGV adoption. In other words, even drastic increases in diesel prices will not lead to greater adoption of alternative technologies, highlighting the superior cost position of combustion engine technologies. Between 2023 and 2044, the scenarios decrease ICEV adoption in favor of LNGVs. The effect is strongest between 2028 and 2035, when sensitivity S2 reduces ICEV adoption by at least 10%. The effect peaks at 12% in 2033. However, ICEVs remain the dominant technology in direct comparison.

Company-internal production of green hydrogen (S3) may induce stark changes to the fleet renewal policy by significantly reducing hydrogen costs. Substantial alterations in adoption can be observed from 2034 onwards. The sensitivity gradually increases FCEV adoption until the end of the time horizon. It also clearly influences only the decision between BEVs and FCEVs. FCEV adoption already surpasses LNGV adoption in 2036 (rather than 2040), and surpasses ICEV adoption in 2040 (rather than 2045). Moreover, FCEVs become the dominant technology in 2041, nine years earlier than in the base model. A clear majority (72%) of the fleet is powered by hydrogen in 2050, representing an 18% increase over the base case.

Sensitivities S6 and S7 concern the FCEV purchase price maturity achieved five years earlier or later. Both scenarios again greatly impact on the adoption decision between BEVs and FCEVs. With earlier price maturity, FCEV adoption rises significantly from 2031 onwards. FCEVs are deployed more often than LNGVs in 2035 (rather than 2040). They also surpass ICEVs and BEVs in 2040 (rather than 2045 and 2050, respectively). In 2050, 62% of the fleet consists of FCEVs, which is 9% higher than the base scenario. Note that the effect is even bigger prior to the final period. Between 2039 and 2048, it is above 10%, with a maximum of 13% around 2040. We see that S6 has a similar effect to S3 (internal hydrogen production). Although the adoption happens a little faster in S6,

the long-term effect on adoption is much higher in S3. Both parameters – the FCEV purchase price and the hydrogen price – are essential for FCEV adoption. The speed of the purchase price decrease seems to carry slightly more weight during earlier periods. Ultimately, the hydrogen price becomes more critical, as it is the main differentiating factor from BEVs in the long run (apart from the nEHS price). On the other hand, a delayed FCEV purchase price maturity (S7) affects adoption shares predominantly after 2045. FCEVs' share of the fleet decreases massively. In 2050, adoption is 19% lower than in the base case (only 35%).

The remaining sensitivity scenarios have only insignificant effects on adoption, and do not noticeably change the average proposed policy. The carbon price scenarios (S4 and S5) only change adoption by 2 to 4% at particular times. Equally, the scenarios for the speed of BEV maturity (S8 and S9) show only 2 to 4% changes in selected phases. The different demand growth scenarios (S10 and S11) alter adoption when the emissions constraint percentage is active, but do not change the average policy (see online appendix for details).

6. Discussion

6.1. Managerial implications

We have shown that our approach can support and facilitate decision making in a fleet renewal setting with alternative technologies. The ADP algorithm enables an efficient, near-optimal solution to the problem and facilitates the formulation of appropriate fleet renewal policies. Furthermore, the generic problem formulation can be easily adapted to various situations in road transportation. For example, it can be used for different fleet types (e.g., light-duty trucks, delivery vans), different operational characteristics (e.g., tour lengths, shift models), or different emissions reduction targets. This would mainly require changing the used values (e.g., consumption, travel distance, distance limits, percentual emissions targets). Additionally, alterations of our assumptions and constraints (e.g., if the purchase budget is irrelevant or BEVs cannot be recharged in the hubs overnight) or the formulation of additional constraints might be necessary (e.g., if not only the distance but transported volumes or weights need to be regarded or fuel station availability is a severe constraint). Studying different competing technologies (e.g., electrofuels) would further require the specification of their modeling (e.g., deterministic, stochastic) and their defining parameters (e.g., development of vehicle purchase, fuel, and infrastructure costs).

Our results suggest that from a cost perspective, moving prematurely to alternative technologies is unwise. Vehicles should instead be held for as long as possible until age or mileage limits demand replacement. This policy reduces the total number of purchases, which largely drive the TCO (especially in short-distance settings). Furthermore, the longer the mobility transition is put on hold, the lower the cost parameters of AFVs become. This leads to a step-function shape in the adoption shares seen in the case study, where BEVs and FCEVs are purchased predominantly when the constraints finally demand them. In addition, identifying clusters with short distances may be beneficial. Effectively, these can be prioritized to enable fast and cost-minimal achievement of the company's aggregate emissions goals. In practice, other factors may need to be considered (e.g., additional tour characteristics, more complex fleet compositions).

The ultimate technology choice depends on the total distance a truck is driven, and should always consider end-to-end CO₂e emissions. Our case study demonstrates that in short-distance operations, it is more difficult for alternative technologies (BEV, FCEV) to compete against combustion engine technologies (ICEV, LNGV). The main reason is that, compared with long-distance operations, fixed costs play a more crucial role than variable costs. Factors such as purchase, infrastructure, and cradle-to-gate carbon costs weigh heavier and delay cost competitiveness. Furthermore, if short-distance transport companies consider only their own emissions, they may move too early and simply shift the emissions upstream. Of course, this may be mitigated if production emissions can be decreased significantly in future. However, even if the latter is achieved, BEVs and FCEVs will still struggle to reach competitive price levels in the next ten to 15 years. Policymakers must investigate external interventions to incentivize a faster mobility transition.

The specific findings of our case study (see Section 5.2) are restricted to companies with similar transport operations. Nevertheless, such companies should carefully monitor the cost components considered. Extreme changes in selected cost types at the upper or lower ends of the modeled stochastic process spectrums may increase the adoption of benefitting technologies by up to 20%. In particular, purchase prices, fuel prices, and the nEHS price may have this effect (see online appendix). In addition, the sensitivity analyses reveal the effect of potential alternative scenarios beyond the stochastic processes of the base model. Significant impacts on adoption are expected from the availability of company-internal green hydrogen production and earlier FCEV purchase price maturity. These two effects each cause FCEVs to be the dominant technology much earlier, and significantly increase the average share of FCEV trucks. On the other hand, later FCEV maturity may drastically decrease average FCEV adoption in the long run. A medium-sized effect should be expected from much higher diesel prices, which may noticeably decrease ICEV and increase LNGV adoption. The carbon price level, BEV purchase price maturity speed, and higher or lower increases in distances driven do not significantly change the dominant fleet renewal policy.

The case study demonstrates that executing the mobility transition requires substantial investment. Fig. 9 illustrates the cumulative increase in the fleet's TCO until 2050. It compares the proposed policy with a (hypothetical) base scenario in which no emissions constraints or regulations become effective, and exclusively ICEVs are used. By 2050, the LSP incurs 4.2% higher TCO when following the optimal fleet renewal policy (on average across all iterations). If evenly spread over all years, this equals an annual TCO markup of 2.9%. Although this policy approximates the cost-optimal fleet renewal with AFVs, the cost rise is still substantial. However, fleet managers should not base their decisions purely on cost criteria. They must also take account of strategic considerations and be mindful of market dynamics. For example, the market requires pioneers who transition to alternative technologies earlier than others. Otherwise, timely achievement of competitive market prices and sufficient market availability of alternative technologies might be at risk. Furthermore, first movers may be able to offer green services at price premiums that may potentially close the cost gaps.

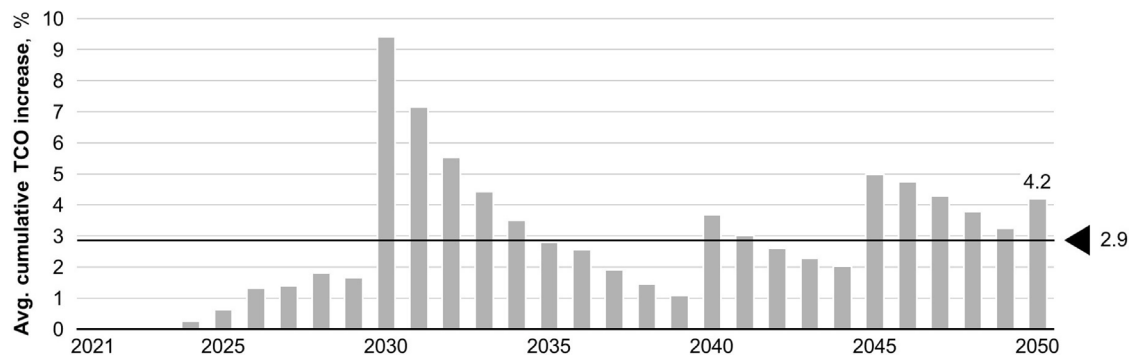


Fig. 9. Cumulative TCO increase of the mobility transition when following the proposed fleet renewal policy.

6.2. Research limitations and outlook

Our study has several limitations. The large scale of the problem requires the selection of approximate optimization methods to solve it in an acceptable time. While ADP is itself an approximate technique, we also use an approximate algorithm for the optimization in every single period. Consequently, the model results are unlikely to describe the best solution among all possible solutions. Although we have proved the high quality of our results, we cannot accurately conclude how far our solution deviates from the best solution. The long time horizon potentially covered by the problem is another limitation to consider. We account for high degrees of fluctuation within the defined stochastic processes. However, some developments are ultimately based on assumptions, which may obviously turn out differently in the future. As a result, we propose to execute the model in an iterative manner. It should be updated with adjusted forecasts and assumptions after a specific time period (e.g., every year, every second year) to continuously inform the fleet renewal strategy while leveraging the latest and most accurate information.

Complementary research can expand decision-making support for practitioners and policymakers. First, an application accounting for different operations characteristics (e.g., long distances), different technologies (e.g., synthetic fuels), or different fleet types (e.g., delivery vans) might provide individual guidance for fleet managers under these kinds of circumstances. Also, complementary selections of stochastic parameters should be examined (e.g., legal requirements, battery production emissions). Investigating different companies' emissions targets might highlight additional meaningful managerial implications. Our results show that technology adoption is driven largely by the timing and size of carbon constraints. For example, the constraints that become effective much later might completely change the fleet renewal policy. It would also be valuable to relax the constraints and investigate how a cost-optimal policy might look. Furthermore, the model might be expanded by including various external interventions, such as political instruments (e.g., CO₂e-driven tolls, funding, technology bans), to examine their effect on LSPs' technology adoption. The results might inform policymakers on the most effective actions to foster the mobility transition.

7. Conclusion

In this paper, we have presented a generic mathematical formulation as a sequential stochastic problem for the case of a transport fleet renewal with alternative technologies. The target function minimizes the TCO of a transport fleet under certain emissions constraints, and can be adapted to various settings. We consider multiple technologies, multiple operational vehicle clusters, and multiple stochastic processes that cover the most critical and uncertain cost types. Thus, our formulation closely mimics the practical problem of mobility transition that all transport companies will soon have to face. This real-world problem quickly becomes large-scale, so many traditional methods cannot be used to derive solutions within an acceptable time. Instead, we propose and develop an ADP approach to approximately solve the problem. This features linear value functions to approximate the future value of a particular fleet portfolio. A dual annealing metaheuristic is used to search for approximate solutions to the optimizations in each time step. Lastly, we apply our approach to a case study of a German LSP operating a fleet of mainly diesel-powered, heavy-duty trucks. In doing so, we prove that our ADP approach is valid and produces high-quality results. We derive valuable managerial implications from the case study that may help practitioners to define their own cost-efficient fleet renewal policies and meet their individual emissions targets.

CRediT authorship contribution statement

Jonas Winkelmann: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stefan Spinler:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Thomas Neukirchen:** Writing – review & editing, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Example calculation of computation time

In the context of the sustainable mobility transition of transport companies, the fleet renewal problem becomes large-scale quickly because long time horizons (e.g., until 2050) and multiple technologies (e.g., ICEV, BEV, LNGV, FCEV) are considered obligatorily. If separate clusters need to be regarded, this adds additional complexity. Imagine we want to determine the optimal number of purchases and the optimal number of sales. Assuming a timeframe of 29 periods (i.e., years) until 2050, four technologies, and four clusters, a total of 928 decision variables has to be modeled. Each variable shall be able to take on values between zero and 49, and the pure computation time for one single combination shall amount to 0.16 s (as measured in our model). When an exact brute force method is applied that calculates all potential combinations and selects the best one, the total required computation time adds up to $2.3 * 10^{1568}$ years. Without considering clusters, total computation time would still amount to $7.6 * 10^{385}$ years. Considering fewer technologies naturally further decreases the computation time, so that three technologies yield $2.2 * 10^{287}$ years, two technologies yield $6.3 * 10^{188}$ years, and a single technology yields $1.8 * 10^{90}$ years. However, this only applies to a deterministic setting where the model must be run only once. In a stochastic setting, the model is executed many times to examine the effects of various random parameter developments. While other exact methods would certainly be faster than the brute force method, they would still reach their limits by facing high numbers of random parameters, time periods, technologies, and clusters. This illustrates that exact algorithms cannot be applied to the underlying problem setting efficiently.

Appendix B. Additional functional dependencies of mathematical formulation

We proceed with a brief explanation of the additional functional dependencies we deliberately omitted in Section 3.2. Please note that we stick to a limited number of potential dependencies that we deem most important. We do not claim to be exhaustive with regards to all potential dependencies between the involved parameters. This has three main reasons. First, the tool is designed to be used iteratively in practice, and assumptions and forecasts are to be updated regularly. An overly complex simulation model striving for perfection in modeling all kinds of cause-and-effect relationships would inhibit its usability. Secondly, it would also require a different methodological approach. We apply an ADP approach with stochastic processes. As opposed to, for example, adopting a system dynamics approach which may capture feedback loops among the variables, we can model only a limited number of interdependencies. And third, many correlations are already indirectly included resulting from how we model the parameters. Parameter developments are informed by reviewed scientific publications. For example, for the random parameters, we fit stochastic processes to specific value corridors taken from matter-specific literature. By doing so, we may model battery and electric vehicle purchase prices independently, but by concluding their values from profound studies that consider their correlated development, we also capture these effects. Furthermore, by additionally integrating these effects into our model, we would risk overestimating them. In our model, the calculations of CO₂e emissions related to vehicle purchases e_{te}^p , operations e_{tevy}^o , and end-of-life activities e_{te}^s are built on a database of various emissions factors. The detailed sources we use to compile the database are found in the online appendix. As a result, CO₂e emissions related to vehicle purchases e_{te}^p and emissions from the end-of-life phase e_{te}^s only depend on the number of vehicles purchased x_{tevy}^p and sold x_{tevy}^s . In contrast, operations emissions e_{tevy}^o depend on the consumption k_{te} , the distance traveled d_{tevy}^d , and the number of vehicles in operation F_{tevy}^a because they combine several emissions factors relating to fuel production and sourcing, transportation activities, and vehicle maintenance. The government funding f_{te} depends on individual funding limits per vehicle and total funding limits per period. We model these limits as deterministic parameters, based on government funding policies passed for upcoming years. While a stochastic formulation would be interesting for the years thereafter, it would also be pure speculation depending on many factors, including political constellations. Instead, our results can support policymakers in finding the right size and timing of potential future funding directives. The number of purchased electric batteries q_{tm}^b is subject to the number of vehicles in operation and their respective mileage (in F_{tevy}^a) as well as the maximum reach of an electric battery l_{te}^b . An electric battery in a selected vehicle is replaced if used over a distance surpassing the maximum reach. The infrastructure cost c_{te}^i depends on the number of vehicles F_{tevy}^a operated in the previous periods. If the infrastructure has been built for a certain number of vehicles of a particular technology before, additional infrastructure might not be necessary. The majority of the stochastic parameters considered in the target function only depend on the period t and sometimes on the technology e . In the model, they are independent of any influence by other parameters. In general, this concerns the purchase price $\hat{p}_{te}^p$, the carbon prices $\hat{p}_t^{cn}$ and $\hat{p}_t^{ce}$, the battery purchase price $\hat{p}_t^b$. In the case study, an exception from this rule is made for the purchase price $\hat{p}_{te}^p$ of two selected technologies e . As BEV and FCEV trucks show large technological similarities, the stochastic processes of their purchase prices are modeled in correlation (see online appendix for details). The energy or fuel prices $\hat{p}_{te}^e$ may depend on multiple variables if certain technologies e are studied. For each fuel type, a base forecast $\hat{p}_{te}^{eb}$ is created by a stochastic process (see Section 4.2). On top of that, final prices might be subject to the German nEHS $\hat{p}_t^{cn}$ and EU ETS $\hat{p}_t^{ce}$ carbon prices. In the case study, we add the price delta between $\hat{p}_t^{cn}$ and the initial price $\hat{p}_0^{cn}$ to the base forecast $\hat{p}_{te}^{eb}$ for diesel, hydrogen, and LNG, proportional to the CO₂e emissions caused during their production and sourcing. We do the same for electricity, applying $\hat{p}_t^{ce}$ instead of $\hat{p}_t^{cn}$. Lastly, the depreciation costs $\hat{c}_{tey}^r$ and the sales price $\hat{p}_{tey}^s$ both not only depend on the period, the technology and the age of the respective vehicle but also the stochastic purchase price $\hat{p}_{te}^p$. We consider when and at which price each vehicle was bought and calculate depreciation and sales prices accordingly.

Appendix C. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.tre.2024.103547>.

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